

the hindu businessline.

SENSEX 74294.96 (-486.80)



IN FOCUS

	Week's close	Week's change
Nifty 50	23346.40	-51.70
P/E Ratio (Sensex)	19.68	-0.13
US Dollar (in ₹)	95.87	+0.32
Gold Std 10 gm (in ₹)	153111.00	+1781
Silver 1 kg (in ₹)	236908.00	+7988

QUANTUM.

Priyansh Mohan's start-up Meine Electric develops iron-air battery system that offers long-duration renewable energy storage **p7**



PULSE.

Medical bodies can now access patient data in exceptional cases, but with caveats **p9**

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QUICKLY.

STRONG PIPELINE

72 firms await SEBI nod to raise ₹1.7 lakh cr via IPOs



Mumbai: The pipeline of initial public offerings (IPOs) is expected to remain robust in the coming months, with 72 companies waiting for SEBI approval to raise about ₹1.7 lakh crore as of August-end, according to data sourced from primedatabase.com. Despite bearish sentiment in the secondary market, it has already been raining IPOs, with 20 companies raising about ₹12,269 crore so far in September. **p2**

TO FIGHT DRONES

DRDO's directed-energy weapon to be ready soon

New Delhi: The Defence Research and Development Organisation's (DRDO) first-of-production model (FOPM) for the 30 kW-class directed-energy weapon (DEW), a strong arm of the 'Sudarshan Chakra' comprehensive air defence system announced by Prime Minister Narendra Modi earlier, will be available within a year to counter swarm drones and unmanned aerial vehicles. **p10**

For India, border peace with China comes before trade ties

CAUTIOUS STANCE. Delhi holding back on big concessions until border stability improves

Amiti Sen
New Delhi

India is not ready to offer big concessions to China on trade and investment until there is greater assurance on peace and stability along the border despite the recent high-level economic engagement between the countries during Chinese President Xi Jinping's visit for the BRICS Summit, sources said.

The position assumes significance following the meeting between Commerce and Industry Minister Piyush Goyal and his Chinese counterpart Wang Wentao on the sidelines of the BRICS Summit in New Delhi earlier this month to resolve long-pending concerns on both sides.

"While the meeting marked a resumption of high-level trade talks, India is not looking at moving ahead on market-access or investment concessions without progress on its concerns relating to the border," a source tracking the matter told *businessline*.

TRADE TALKS

Xi visited India this month after seven years as ties had



STRATEGIC RESTRAINT. A file photo of Commerce Minister Piyush Goyal with his Chinese counterpart Wang Wentao

soured following the 2020 Galwan Valley clash, leading to prolonged tensions along the Line of Actual Control (LAC). The visit, on September 12, came amid renewed efforts by India and China to ease tensions, although recent disputes over alleged Chinese encroachment have reinforced India's priority of maintaining peace and stability along the border.

At the meeting between the two trade Ministers, India raised China's restrictions on exports of rare earth minerals and high-technology products, besides flagging the large and widening trade deficit.

"Beijing did not provide specific assurances on these issues," another official said.

While bilateral trade between India and China touched a record \$151 billion in FY26, the balance remains heavily skewed in China's favour, with imports from China into India at \$131.6 billion.

CHINA'S CONCERN

China, meanwhile, raised India's restrictions on investments from Chinese companies and sought greater liberalisation of the investment regime, the official added.

FDI from countries sharing a land border with India, including China, require prior government permission, which could take a long time.

"Although some conces-

sions were made for investments coming in from border countries, including China, earlier this year, it was nominal and Beijing wants the investment restrictions to be totally removed. But there is not much scope for anything substantial happening now till Beijing contributes to improving the border situation," the first source said.

At their bilateral meeting on the sidelines of the BRICS Summit, Prime Minister Narendra Modi told Chinese President Xi Jinping that "peace and tranquillity" in the border areas remained an "essential basis" for the continued development of bilateral relations.

He also stressed the need to observe existing agreements and understandings on border-related issues.

Xi, on the other hand, called for the two countries to take a strategic and long-term view of their relationship, cooperate for common development and advance bilateral relations alongside efforts to resolve the boundary question. China's readout called for "parallel, mutually reinforcing progress" on the two fronts.

MoSPI plans to release pre-FY23 GDP back series by year-end

Shishir Sinha
New Delhi

With the change in base year to 2022-23 from 2011-12, India plans to release back series GDP data for years prior to 2022-23 by the end of this year, said MoSPI Secretary Saurabh Garg.

"Releasing two official growths of GDP by running the old and new series concurrently would add to more ambiguity and confusion among the users and, therefore, none of the countries run the parallel series together," he told *businessline*.

Interview p3

SEBI board may discuss PMS, settlement norms

Akshata Gorde
Mumbai

The SEBI board, which is scheduled to meet on September 24, may consider an overhaul of settlement regulations and portfolio management services (PMS), a common advertisement code for regulated entities and allowing FPIs access to commodity derivatives.

Details p2

Tata Trusts cites SC's Cyrus Mistry ruling to counter Chandra

Our Bureau
Mumbai

The challenge mounted by the Tata Trusts to the re-appointment of N Chandrasekaran as Chairman of Tata Sons draws its strength from the Supreme Court's ruling in the Cyrus Mistry case, where the court upheld the sanctity of the Tata Sons' Articles of Association (AoA) and recognised the special rights accorded to Tata Trusts as the majority shareholder.

In a statement on Sunday, the Trusts argued that the September 17 board resolution granting Chandrasekaran a fresh five-year term failed to satisfy a mandatory requirement under the AoA.

According to the Trusts, certain decisions of the Tata Sons board require the affirmative support of a majority of Tata Trusts-nominated directors. With one of the two Trust nominees voting against the resolution, that condition was not met, rendering the resolution invalid regardless of the overall board vote.

The Trusts maintained that the Chairman's casting vote could not be used to overcome this deficiency. While a casting vote may

break a tie in the overall board vote, it could not override a separate affirmative voting requirement that is explicitly embedded in the company's constitution. "The Board put a question, and the AoA answered it in the negative," the Trusts said, arguing that the resolution was void *ab initio* (void from the outset).

LEGAL BATTLE

The Trusts' argument is closely tied to the position adopted by Tata Sons during the legal battle stemming from Mistry's removal as Chairman. In those proceedings, Articles 104B and 121, which confer affirmative voting rights on Tata Trusts-nominated directors, were challenged as oppressive.

Tata Sons strongly defended those provisions, arguing that they were legitimate protections available to the majority shareholder rather than instruments of oppression. The Supreme Court ultimately endorsed that view and upheld the validity of the Articles.

The Trusts now contend that Tata Sons cannot depart from an interpretation of the AoA that it successfully persuaded the country's highest court to uphold.

Also read p10

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QUICKLY.

BEML bags ₹5,400 cr order for 16 high-speed trains

Ahmedabad: BEML Ltd has secured a ₹5,400 crore contract from the National High Speed Rail Corporation Ltd to supply and maintain 16 high-speed trains for the Mumbai-Ahmedabad High Speed Rail corridor, marking the largest single order in the company's history and taking its overall order book beyond ₹21,000 crore. These train-sets have to be delivered in the next seven-eight years, official sources told *businessline*. **OUR BUREAU**

Mother Dairy sets up dedicated butter chiplet

Ahmedabad: Mother Dairy has set up its first dedicated butter chiplet manufacturing line in Junagadh, Gujarat, with an annual production of 550 tonnes, as the firm looks to expand its presence in the butter segment. The new line has been established at Mother Dairy's existing milk processing facility in Junagadh and will supply butter chiplets across India, a company official said on Saturday. **OUR BUREAU**

SEBI board to consider settlement, PMS, FPI rule changes on Sept 24

ON THE AGENDA. FPI participation in commodity derivatives, advertising framework may also be discussed

Akshata Gorde
Mumbai

The SEBI board is scheduled to meet on September 24 and is expected consider an overhaul of settlement regulations and portfolio management services (PMS), a common advertisement code for regulated entities and greater access for foreign portfolio investors (FPIs) to commodity derivatives, according to people familiar with the matter.

The board may also take up proposals relating to depository receipts for REITs and InvITs, expansion of the regulatory framework for vault managers and changes to debt market rules, among others.

MF PORTFOLIOS

The markets regulator has proposed a comprehensive review of regulations governing portfolio managers,

KEY PROPOSALS

- Settlement regulations may see a comprehensive overhaul
- PMS regulatory framework could be revamped
- Wider FPI participation in non-agri commodity derivatives proposed
- Common ad code, REIT/InvIT DRs and vault manager changes may also be considered

including allowing mutual-fund-only PMS wherein portfolio managers could offer professionally managed portfolios comprising mutual fund schemes.

It has also suggested widening the investment universe available to PMS firms, including permitting investments in yet-to-be-listed securities and allowing discretionary portfolio managers to invest up to 10 per cent of a client's assets under

management in investment-grade unlisted debt securities.

SEBI has also proposed an overhaul of the settlement framework, including changes aimed at rationalising settlement amounts and speeding up smaller cases.

The proposals also seek to allow settlement at later stages of proceedings, including when a matter is before the Securities Appellate

Tribunal or the Supreme Court, and introduce a fast-track route for specified violations and cases involving settlement amounts of up to ₹10 lakh.

Other changes include issuing settlement notices before show-cause notices in certain cases, extending the time to apply for settlement after a show-cause notice from 60 days to 90 days and reducing the additional amount payable for refiling a withdrawn settlement application from 50 per cent to 20 per cent.

FPI EXPANSION

Another proposal seeks to widen FPI participation in exchange-traded commodity derivatives. SEBI has proposed allowing FPIs to participate in non-agricultural commodity index derivatives and in non-agricultural commodity derivative contracts that are settled through physical delivery,

subject to prescribed conditions.

The board may also consider SEBI's proposed common advertisement code for regulated entities, which seeks to bring advertising requirements for different regulated entities under a common framework and includes provisions allowing celebrity endorsements by regulated entities subject to conditions.

The regulator has also proposed expanding the scope of its Vault Managers Regulations, 2021, which currently govern vault managers providing services for electronic gold receipts traded on recognised stock exchanges.

Separately, SEBI has proposed enabling depository receipts against units of REITs and publicly listed InvITs, which will allow these units to be accessed by investors through overseas markets.

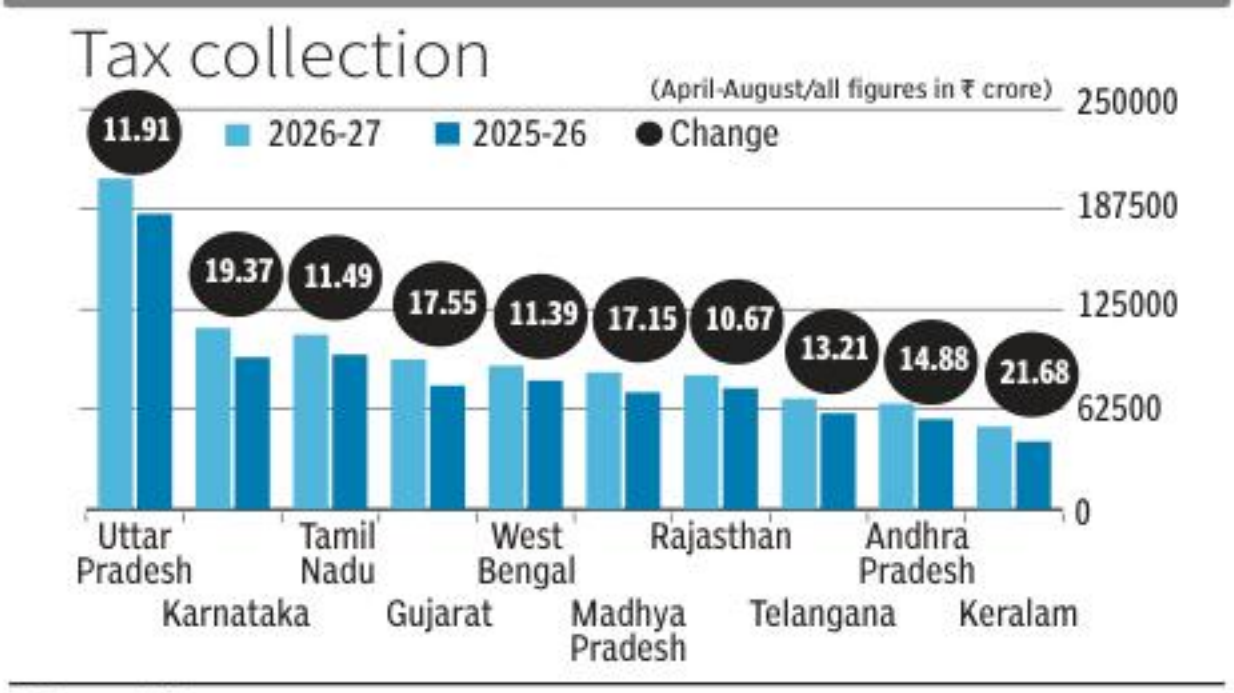
U.P. affirms top position in tax collection, followed by Karnataka, T.N.

Shishir Sinha
New Delhi

Uttar Pradesh continues to lead the States and Union Territories in collection from taxes during the April-August period of the current fiscal, followed by Karnataka and Tamil Nadu, data from the Comptroller & Auditor General (CAG) show.

The State also tops in terms of the highest number of GST assesses with over 22 lakh. Uttar Pradesh collected over ₹2.06 lakh crore during the five-month period, is around 12 per cent higher than the corresponding period of the last fiscal.

Keralam, though last in absolute numbers among 10 States (see table), recorded



the highest growth at around 22 per cent. The study does not include Maharashtra, as five-month data for the said State is not available.

GST GROWTH

The States' tax revenue comprises seven components — State Goods & Services Tax

(SGST), stamps & registration, land revenue, sales tax, State excise duty, the State's share of union taxes and other taxes & duties.

Among these, SGST has the maximum share. All the 10 States taken for this report have shown collection from GST increasing despite

rate rationalisation and doing away with compensation. For example, according to data compiled by the CAG, Uttar Pradesh earned over ₹73,000 crore through SGST during the April-July period of the current fiscal as against over ₹66,000 crore during the corresponding period of last fiscal.

In terms of capital expenditure, Gujarat was ahead of the others with spending of over ₹29,600 crore in the first five months of the current fiscal as against around ₹26,000 crore during the corresponding period.

At the same time, while Uttar Pradesh spent over ₹21,000 crore during the five-month period, like the corresponding period of the

last fiscal, the amount for Karnataka was over ₹15,600 crore against around ₹14,000 crore.

All these data are being studied at a time when the Centre has convened a two-day conference over the weekend with the States and the Union Territories (with legislatures) on "Financing India's Journey towards Viksit Bharat."

CONFERENCE TALK

During the conference, while Economic Affairs Secretary Anuradha Thakur emphasised the need for higher private finance, Chairman of the 15th Finance Commission NK Singh expanded the deliberations by suggesting measures for revenue augmentation by the States, besides

issues such as debt-GDP ratio and going from public finance to private capital, among others.

According to Thakur, keeping in mind the ongoing global difficulties and because of the scale of transformation, "We envisage ourselves to reach our shared goal. It needs to be borne in mind that this scale cannot be met solely by government budgets, and that private sector financing will need to play a critical role."

Suggesting ways for how the States could boost revenue, he said they were better placed to leverage their local information advantage. The opportunity, therefore, is to turn this information advantage into stronger revenue mobilisation.

Stellantis takes full ownership of Tiruvallur plant, buys out stake of CK Birla Group

T E Raja Simhan
Chennai

Stellantis India has taken full ownership of its Tiruvallur plant by buying the stake of Hindustan Motor Finance Corporation Ltd (HMFCL), a CK Birla Group company, in the joint venture entity Stellantis Automobiles India Pvt Ltd (SAIPL).

The details of the amount invested in the transaction and the percentage of stake bought were not disclosed.

The acquisition would provide greater flexibility and agility in decision-making and enable the company to respond faster to market and customer requirements, Shailesh Hazela, CEO and Managing Director, Stellantis India, told *businessline*.

"Both partners agreed to keep the information on

equity sold and amount paid confidential," he added. The firm is working with the Tamil Nadu government for additional investment linked to the production ramp-up and it will be announced, Hazela said.

Stellantis and the CK Birla Group began their manufacturing and mobility partnership in 2017. The Tiruvallur facility currently manufactures the Citroën C3, ë-C3, C3 Aircross and Basalt. The plant has achieved more than 95 per cent localisation, per Hazela.

"Stellantis has invested over €1 billion in India to support its long-term growth ambitions, including manufacturing, product development, localisation and capability building. The Tiruvallur facility benefits from a strong supplier and logistics ecosystem while



Shailesh Hazela, CEO and MD, Stellantis India

contributing to Tamil Nadu's position as a leading automotive manufacturing hub," he said.

FUTURE LOOKOUT

Stellantis is targeting to produce over 50,000 units annually by 2027 from 18,000 units in 2026, which is also expected to increase employment at the facility. Currently,

around 70 per cent of the plant's production is for the domestic market and 30 per cent for exports. By 2027, Stellantis expects this mix to shift to roughly 40 per cent domestic and 60 per cent exports. The Tiruvallur facility exports vehicles to South Africa, Egypt, Indonesia, Nepal, Bhutan and Sri Lanka and is evaluating additional markets, he said.

The plant currently has capacity of around 60,000 vehicles, with further headroom available through additional investment and higher utilisation, he added.

Stellantis also has an engine and transmission facility at Hosur, which produces around 2,50,000 engines and 3,50,000 transmissions annually. This supplies not only to the Tiruvallur plant but also to Europe, the Middle East and Africa, he said.

Grahaa Space accelerates launch plans for Earth-observation satellites

Aishwarya Kumar
Bengaluru

Bengaluru-based Grahaa Space is stepping up its launch cadence after successfully putting its SOLARAS satellite into orbit aboard Skyroot's Vikram-1, targeting two technology-demonstration missions in Q4 FY26 and Q1 FY27 and potentially three launches by the second half of FY27.

The start-up has already signed two customers for its upcoming missions, Co-founder and CEO Ramesh Kumar said in an interview.

The missions will help Grahaa validate its optical payload, a key step before it begins building its planned constellation of nano satellites to deliver near-real-time video data of the Earth.

"The next mission is proving our optical payload. It is



Ramesh Kumar V, Co-founder & CEO, Grahaa Space

the second technology demonstration mission," Kumar said, adding that once the payload is proven, "we will start building the constellation. There we will sell the data."

EARLIER FAILURE

The company's push to accelerate its launch cadence comes after it lost an earlier satellite in a launch failure.

72 companies await SEBI nod to raise ₹1.7 lakh cr via IPOs

Suresh P Iyengar
Mumbai

The pipeline of initial public offerings (IPOs) is expected to remain robust in the coming months, with 72 companies waiting for SEBI approval to raise about ₹1.7 lakh crore as of August-end, according to data sourced from primedatabase.com.

Some of the companies awaiting SEBI approval for DRHPs include Mahanadi Coalfields (₹10,000 crore), Carlsberg India (₹6,300 crore), Sembcorp Green Infra (₹3,750 crore), Emerald Jewel Industry India (₹3,500 crore), Encube Ethicals (₹3,000 crore) and Veritas Finance (₹2,800 crore), it added.

Despite bearish sentiment in the secondary market, it has been raining IPOs, with 20 companies raising about ₹12,269 crore so far in September.

GATHERING PACE

In fact, a record six companies listed on a single day last week after attracting demand of ₹1.4 lakh crore against an issue size of ₹7,100 crore. The rush of IPOs is expected to gather further pace this month, with SEBI extending the validity of approvals for IPOs and rights issues expiring between April 1 and September 30, 2026, to September 30, 2026.

VK Vijayakumar, Chief Investment Strategist, Geojit Investments, said there is an element of irrational exuberance in the booming IPO market as most retail investors are bidding for listing gains and, therefore, are not giving due importance to valuations and management quality. As the average returns from IPOs during the period from early June to September 7 were 22 per cent, retail investors have been chasing IPOs, he added.

THE REAL CHALLENGE

Sandeep Gupta, Director, Valmiki Leela Capital, said

Waiting for SEBI approval

	Issue size (₹ cr)
Cult.Fit	3,500
Tablespace Technologies	3,350
Svatantra Microfin	3,000
Muthoot Fincorp	3,000
Encube Ethicals	3,000
Veritas Finance	2,800
Sagar Defence Engineering	2,000
Atomberg Technologies	1,500
Assetz	1,200

Source: primedatabase.com

the real test for IPOs in the pipeline will be the willingness of issuers to launch issues at valuations that the market could absorb, and this could prompt some companies to wait for more supportive conditions.

Several 2026 listings are trading below their issue prices, showing that listing gains can no longer be taken for granted. Retail investors are increasingly paying attention to post-listing performance, which is putting greater emphasis on valuation discipline, he added.

"Jio Platforms will be closely watched, given its proposed fresh issue with no offer for sale. A well-received issue at a valuation that investors find attractive will reinforce confidence across the IPO pipeline," he said.

Anoosha Soham Bathwal, CEO & Founder of Dhanvestor, said there had been immense liquidity and enthusiasm in the market, especially from retail investors looking for a stake in India's success story. However, some weak listing performances by recently listed companies had provided a reality check for investors, she added.

"Investors should evaluate the business model, earnings predictability, competitive edge and, especially, valuation relative to peers. Even an excellent company can become overvalued once all future growth is priced in," said Bathwal.

P4 poverty initiative gains ground in A.P.

G Naga Sridhar
Hyderabad

The flagship poverty alleviation programme of Andhra Pradesh, Public-Private-People Partnership (P4), is steadily expanding its reach.

According to the latest government data, Palnadu district has recorded 98.61 per cent integration of village/ward secretariats with Bangaru Kutumbams (BKs), placing it among the districts with the highest level of linkage.

Palnadu is followed closely by Parvathipuram Manyam at 98 per cent, Visakhapatnam at 98.37 per cent and Guntur at 98.32 per cent. Anakapalli recorded 97 per cent integration, followed by NTR district at 97.69 per cent, Tirupati at 97.36 per cent, Nellore at 96 per cent and Srikakulam at 96.43 per cent.

The data indicate that several districts have achieved integration levels of more than 95 per cent. At the lower end, YSR Kadapa recorded 72.22 per cent, while Sri Sathya Sai stood at 76 per cent. Dr BR Ambedkar Konaseema at 79 per cent, Alluri Sitharama Raju at 80 per cent and Anantapur at 82 per cent.

A brainchild of Chief Minister N Chandrababu Naidu, the programme aims to bridge the socio-economic divide by encouraging the wealthiest 10 per cent of the population to support the bottom 20 per cent of the poorest families.

Adani Power lays foundation stone for ₹48,000 crore thermal project in Assam

Our Bureau
Ahmedabad

The foundation stone for Adani Power's 3,200 MW thermal power project was laid in Assam on Sunday, marking an investment of around ₹48,000 crore and setting the stage for what the company says will be the "North-East's largest power generation asset".

The project is being developed at Chapar in Assam's Dhubri district. Speaking at the foundation stone ceremony, Jeet Adani, Director, Adani Group, said the company had already conducted four to five job drives in Assam, with around 4,500 local youth hired for the project that will bring together thousands of workers, engineers, technicians, contractors and suppliers.



Jeet Adani, Director, Adani Group

The 3,200 MW project marks a significant expansion of Adani Power's presence in Assam, and Adani said the group's businesses in Assam are spread across sectors including power, airports, smart meters, cement, connectivity and maintenance, repair and overhaul services.

'BATTERY OF INDIA'
"Assam has the potential to become the 'Battery of India'," Adani said, outlining

the group's long-term view of the State's energy potential.

The foundation stone ceremony comes against the backdrop of the Adani Group's broader investment plans for the North-East.

At the Rising Northeast Global Investors Summit last year, Group Chairman Gautam Adani had announced plans to invest up to ₹1 lakh crore in the North-East over the next decade.

Jeet Adani said work was already underway in Assam with investments of nearly ₹80,000 crore.

Beyond power generation, the Group is also developing and operating Guwahati's Lokpriya Gopinath Bordoloi International Airport, where it plans to develop an airport city with commercial, hospitality, retail and entertainment infrastructure.

Core tech, engineering services lift India's FY26 IT export growth despite software product slowdown

Sindhu Hariharan
Chennai

India's software exports growth is being led by traditional information technology (IT) services and product design and engineering services, while the software product development and BPO services are seeing a slowdown, according to the Reserve Bank of India's latest survey on computer software and IT-enabled services exports for FY26.

SERVICES SURGE

Total software services exports rose from \$204.7 billion in FY25 to \$221.4 billion in FY26, an increase of 8.2 per cent.

This was driven by a 12 per cent and an 11 per cent

Engineering services power India's IT exports (in \$ billion)			
	FY 2025	FY 2026	Growth %
IT services	131.3	147.0	12.0
Software product development	6.8	6.4	-5.9
BPO services	55.8	56.0	0.4
Engineering services	10.8	12.0	11.1
Total software services	204.7	221.4	8.2

Source: RBI

growth respectively, in IT services and engineering services. The share of IT services in total exports rose sharply from 64.1 per cent in FY25 to 66.4 per cent of total software exports in FY26, a trend that analysts attributed to rising GCCs making up for the revenue slowdown of traditional IT firms.

In contrast, the share of software product development fell from 3.3 per cent to 2.9 per cent, considering the

maturity of AI. Within engineering services, the segment called 'product design engineering' (mechanical electronics, excluding software) saw a sharp rise in its share from 2.7 per cent to 4.3 per cent, suggesting that India is maturing as a hardware product engineering hub.

Further, embedded solutions' exports grew in share from 2.7 per cent to 3 per cent. This refers to the export of specialised engineer-

ing, design and software services towards building embedded systems.

SOFTWARE EXPORT

Software exports to the US increased from \$108.3 billion in FY25 to \$119.7 billion, with the share of exports to the US rising from 52.9 per cent to 54.1 per cent.

In contrast, Europe's share slipped by one percentage point, despite exports to the region rising in absolute terms.

The RBI's survey also showed that the US dollar was the principal invoicing currency for India's software exports in the fiscal year, followed by the euro, the rupee, and the pound sterling.

In India, software services exports are primarily delivered through the off-site mode, which holds a 91.7 per cent share in FY26.

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Release of pre-FY23 GDP back series likely by year-end: MoSPI Secretary

DATA RESET. Saurabh Garg discusses the fine print of the base year revision of the GDP

bl.interview

Shishir Sinha
New Delhi

The government is planning to release back series for years prior to 2022-23 by the end of this year, said Saurabh Garg, Secretary in the Ministry of Statistics and Programme Implementation (MoSPI). He discussed the fine print of the base year revision of the GDP.

Edited excerpts:

The key issue in the GDP debate is the use of double deflation and the use of producer/input prices. What new datasets has MoSPI obtained that make these methods feasible?

The transition to double deflation was not sudden. It was built on extensive experiments and detailed analysis using historical time-series data from Annual Survey of Industries (ASI) spanning the period 2011-12 to 2023-24, supply and use tables (SUT) and the WPI of the 2011-12 series under the guidance of the Advisory Committee on National Accounts Statistics (ACNAS). The stability of output PPI from 2022-23 onwards was assessed and MoSPI examined the feasibility of using output PPI in national accounts prior to its official launch.

The DPIIT initiated the exercise for compilation of the Output Producer Price Index and hence, the base revision of GDP was an opportune time to introduce double deflation. Since output PPI was not released by February 27, double deflation was carried out using available item-level WPI (2011-12 series). Later on, the release of output PPI in June 2026, following out-

put PPI-based double deflation, was implemented and updated estimates were released on August 31, 2026.

What are the sector-wise price indices and weights producing the aggregate deflator? Given the big divergence between the GDP deflator and WPI/CPI, can MoSPI provide a reconciliation showing why the GDP deflator behaves so differently?

In the new base year, the deflation strategy in various sectors has been volume extrapolation, single extrapolation and double deflation; use of item-specific price indices has been emphasised in the new base compared to aggregate price indicators applied at the sectoral level.

Since in the old series on applying single deflation (the output and input are deflated by the same deflator), the movement in IPD (Implicit Price Deflator, commonly called the GDP deflator) of the National Accounts Statistics was close to the WPI/CPI. However, in the new series, the approaches used in constant price estimation are more sophisticated and hence, do not have a one-to-one correspondence with output PPI and CPI.

Under the new national accounts methodology, manufacturing uses double deflation: Output and intermediate inputs are separately deflated. Therefore, manufacturing GVA's implicit price change need not equal manufacturing PPI/WPI inflation when the output and input prices vary differently, and this fact is addressed in double deflation.

Similarly, CPI reflects price changes of a specific basket curated based on the household consumption of goods and services, whereas

In this new series, several methodological improvements have taken place. However, the major factor is mainly the new data for estimating the unincorporated sector

SAURABH GARG,
Secretary, MoSPI



GDP by demand-side covers government consumption, investment, export and import too, for which altogether separate price deflators have been adopted.

The GDP deflator is economy-wide: It captures price changes across all domestically produced final goods and services. WPI, by contrast, is a wholesale-price index for goods, with no comprehensive coverage of services. MoSPI explicitly notes that the GDP deflator need not move in line with WPI/PPI/CPI because of differences in their coverage and use of item-level price indices.

Why was the new methodology not run in parallel with the old one before it became the basis for the official series?

Several data sources, such as ASUSE, PLFS and output PPI, apart from methodological changes such as segregation of GVA of multi-activity enterprises, have been introduced in an integrated fashion in the new series. Bifurcation of growth in GVA to the one due to double deflation and that due to new data sources, coverage and other methodological changes is not feasible. Further, releasing two official growths of GDP by running the old and

new series concurrently would add to more ambiguity and confusion among users and, therefore none of the countries run the parallel series together. India is planning to release the back series for years prior to 2022-23 by the end of this year.

Can MoSPI demonstrate that the improvement is not merely a change in methodology but also an improvement in the quality and availability of the underlying data?

In earlier base series (e.g., 2011-12), the GVA of the unincorporated sector relied heavily on the Enterprise Survey conducted in the base year and the quinquennial round of the Employment Unemployment Survey. In subsequent years, the benchmark/base year estimates were extrapolated using relevant proxy indicators, such as growth in sales tax for extrapolating the base year GVA of trade. MoSPI now conducts the Annual Survey of Unincorporated Sector Enterprises (ASUSE), alongside the Periodic Labour Force Survey for labour force estimation on a continuous basis. Thus, the use of proxy indicators is not required.

Then, PFMS for various States were not available in the previous base. GST,

which is one of the important administrative data sources for quarterly GDP estimation, was launched in 2017-18. Eventually, the GSTN database has strengthened substantially.

Also, reconciling production-side GVA and expenditure-side GDP often resulted in sizeable discrepancies due to asymmetric data availability. In the new series, the compilation of GDP by production and expenditure approaches has been integrated with the SUT framework.

From ₹86 lakh crore to ₹80 lakh crore is a massive downward revision in the 2026 nominal GDP, and that has been questioned. What do you say?

Our nominal GDP in the new series has been revised down by about 2.7 per cent in 2022-23, 3.5 per cent in 2023-24 and 3.8 per cent during 2024-25. Such changes in base revision are due to the incorporation of better data sources and improved methodology. In this new series, several methodological improvements have taken place. However, the major factor, taking the levels down, is mainly the new data for estimating the unincorporated sector. Results of ASUSE and PLFS data provide a more direct basis for measurement of unincorporated sector contribution to value added on a regular basis in the new base year.

However, a proper assessment of the sector-wise picture shows revisions have happened both upward and downward. Agriculture and allied activities GVA were revised up by about 3.8-5.9 per cent; financial services and real estate, professional services and ownership of dwellings were revised up by approximately 7.8-9 per cent over the comparable years.

OMCs' under recovery on LPG up 5% in Aug

Rishi Ranjan Kala
New Delhi

The under recovery on LPG of PSU oil marketing companies (OMCs) rose roughly 5 per cent on a monthly basis to more than ₹62,000 crore in August 2026 as retailers continue to sell the critical cooking fuel at below the market rates.

The under-recovery was around ₹721 per cylinder in June, which fell to roughly ₹500 a cylinder a month later. The under recovery per cylinder stood about ₹188 in August, which has risen to roughly ₹210 a cylinder in the current month.

The cumulative under recovery of PSU OMCs rose from roughly ₹51,000 crore in June to over ₹62,000 crore last month. PSU OMC losses ballooned after they sold LPG below international prices, which surged significantly after February 28, due to the West Asia conflict and the closure of the Strait of Hormuz (SoH).

Prior to the outbreak of the conflict, India used to import about 60 per cent of its LPG consumption, of which about 90 per cent transited through the SoH.



LPG imports rose to a six-month high of 1.3 million tonnes (mt) in August, growing by roughly 50 per cent month-on-month. However, the cargoes were down by almost 33 per cent on an annual basis.

A back of the envelope calculation shows that for \$1 million, India procured roughly 1.65 tonnes of LPG in August 2026, roughly the same rate during March 2026. During April, around 1.39 tonnes of LPG was procured for \$1 million.

SUPPLY DISRUPTIONS However, the quantity fell as supply disruptions in the SoH completely choked off LPG supply during May and June.

For instance, India paid \$1 million to procure around 1.17 tonnes and 1.18 tonnes

of LPG in May and June, respectively.

The US was India's top supplier last month. India imported about 0.62 mt LPG from the US in August, after bringing in 0.89 mt in July, OPEC data showed.

Washington, which was India's fifth largest LPG supplier till January 2026, jumped a spot to become the fourth largest, replacing Kuwait a month later.

The fresh conflict in West Asia (from February 28) propelled the North American country to become the top supplier for five consecutive months, beginning March. However, this diversification has come at a cost.

The US LPG is linked to Mont Belvieu prices (\$0.68 per gallon in August 2026, equivalent to around \$360 per tonne) which is generally benched lower than Saudi CRP, but the longer voyages to India raise its delivered cost, Equirus Securities pointed out.

Meanwhile, constrained regional supply pushed Saudi Arabia's September propane contract price up by \$5 per tonne to \$625. Higher sourcing costs are expected to continue to accumulate with Indian OMCs, it noted.

TODAY'S PICK.

KPR Mill (₹1,141.30): BUY

Akhil Nallamuthu
bl. research bureau

The stock of KPR Mill has been in an uptrend since April. Back then, it found support at ₹800 and established an uptrend.

Recently though, it has been witnessing higher volatility, particularly since the second half of June. While the trend has not turned bearish, the direction of the movement was largely sideways. But now it has found a support at ₹1,100, where a key rising trendline coincides. The rebound on Friday indicates it might gain more strength going forward.

We see a potential for the stock to appreciate to ₹1,400 in

the medium-term. So, buy now at ₹1,141 and ₹1,100. Place the stop-loss at ₹1,000. When the stock rallies to ₹1,250 and ₹1,340, alter the stop-loss to ₹1,190 and ₹1,300, respectively. Book profits at ₹1,400.

Note: The recommendations are based on technical analysis. There is a risk of loss in trading

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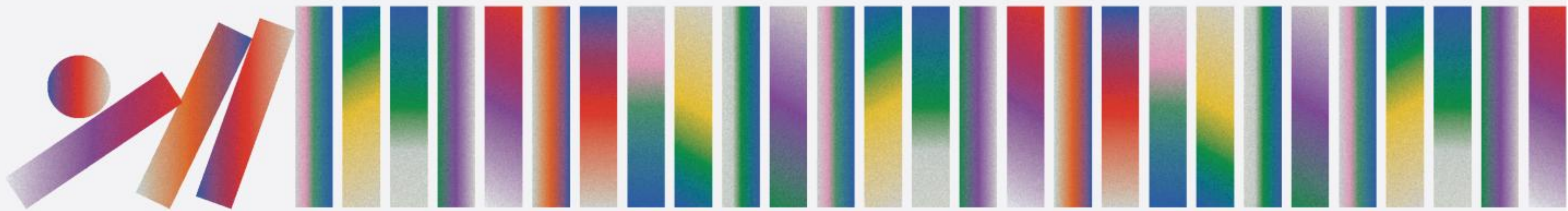
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Stick it out

Tariff threat over Russian oil shouldn't rattle India

The Lindsey O Graham Sanctioning Russia and Iran Act of 2026, which came into effect last week, empowers President Trump to impose up to 100 per cent tariff on the top five buyers of Russian energy products, if the countries concerned do not back down and satisfy the US that they will stop buying from Russia.



The Ministry of External Affairs has responded well by saying that India will secure its 'trade and economic interests' and source its needs accordingly. The Trump administration is nothing if it is not whimsical. It may seek to leverage concessions for the trade pact, as a *quid pro quo* for relaxing a law that gives the President powers to relax its penal provisions. However, Trump can go back on his word, as he has done with countries that have inked deals with the US. Therefore, the prospect of dealing with a 100 per cent hike need not influence India's decisions now. There are also reasons to believe that the law, harsh as it appears, may not be strongly enforced. Trump has advised Ukraine not to attack Russian refineries for fear of causing a shortage and 'hurting the world'. So, the US wants to squeeze Russian supplies, but not by so much as to send crude prices spiralling over current levels of around \$100 a barrel.

The US would surely be aware that rising prices will spur shadow shipments of discounted Russian oil. These have continued, despite Western efforts to stop them. The law may come up for review after 180 days, and India can wait it out. At the outset, it puts India and China in the line of fire for being among the top five importers of many energy products. According to the Centre for Research on Energy and Clean Air, the top buyers of Russian oil since the Ukraine war broke out in 2022 are China (about 50 per cent), India (37 per cent), Turkiye (5 per cent) and EU (5 per cent). In the case of LNG, however, EU accounts for 49 per cent of Russian exports, followed by China (24 per cent), Japan (18 per cent) and South Korea (6 per cent). China (37 per cent), India (19 per cent), Turkiye (15 per cent), South Korea (12 per cent) and Vietnam (4 per cent) top the list of top buyers of Russian coal. For natural gas, the penal tariff will apply on countries that account for more than 15 per cent of Russia's gas exports over the preceding year.

This possibly lets off the EU, whose share in Russia's exports of LNG has fallen. For other importing categories, the law applies if countries 'knowingly' make purchases of Russian energy 30 days following the enactment, and have been among the top five importers over the last year. However, China is too powerful to be openly targeted. India's purchases of Russian crude have dipped from July's high of 2.8 million barrels per day (51 per cent of its total crude imports), to 2.1 million bpd in August and 1.4 million bpd so far in September, according to private data sources such as Kpler. A declining trend may not entitle India to a reprieve. India should be pragmatic and pursue all trade options.

OTHER VOICES.

THE WALL STREET JOURNAL.

EUROPE

The Saudis, the Houthis, and Israel

Is the Houthi crisis a diplomatic opportunity? Saudi Arabia has come running to Washington and even to Israel in its hour of need. Meanwhile, Riyadh's feud with the United Arab Emirates continues to backfire, and the safeguard-breaking Saudi nuclear deal with President Trump has been submitted to Congress. With all of these moving pieces, there may be a checkmate for Mr. Trump to find. The point isn't that he must answer the calls of Saudi Crown Prince Mohammed bin Salman for U.S. airstrikes. Instead the Houthi blockade on Saudi oil has already become the impetus for the kind of Saudi-Israeli-Emirati cooperation that only U.S. Central Command can convene. That's the regional alliance needed to counter Iran and its Houthi proxy militia. (NEW YORK, SEPTEMBER 16)

讀賣新聞

THE YOMIURI SHIMBUN

Central bank should not be slow to act

The risk is growing that the inflation rate will overshoot the target due to soaring crude oil prices and a weak yen. It is crucial for the Bank of Japan to decide on additional rate hikes at an appropriate time to curb inflation. The BOJ decided to raise the policy interest rate by a quarter percentage point to around 1.25%, bringing it to a level not seen in about 31 years, since 1995. After ending its large-scale monetary easing policy in March 2024, the central bank raised the policy rate roughly once every six months. The latest rate increase follows the previous one by just three months, the shortest interval since 2024. The BOJ has so far adopted an accommodative monetary policy to boost the underlying inflation rate. (TOKYO, SEPTEMBER 19)



TC SUSEEL KUMAR
R SUDHAKAR

A fixed deposit matures in five years; an equity share can be sold tomorrow. A whole life policy taken out by a father naming his newborn son may not discharge its obligation for 70 years. An immediate annuity bought at 30 pays as long as the annuitant lives — beyond a 100, if he lives that long. These are not rare cases. They are the product. And they change what regulation must do.

Most people assume insurance regulation exists to protect policyholders from bad companies — fraud, mismanagement, insolvency. That is incomplete. It exists to protect the contract from the mortality of companies. Companies fail, merge, restructure and exit markets. The contract can do none of these things; it must survive whatever happens to the institution that wrote it. That is the organising principle behind every provision governing mergers, amalgamations, portfolio transfers, run-offs and winding-up in life insurance law, in India and abroad.

WHAT THE INDIAN LAW ENCODES

The Insurance Act, 1938 addresses this in a cluster of provisions often cited but rarely explained. Under Section 35, no life insurance business may be transferred or amalgamated with that of another insurer except under a scheme approved by IRDAI. Section 36 goes further: before approving one, IRDAI must send notice to every life policyholder concerned and hear any who applies to be heard. The policyholder is not a passive party to a corporate transaction but a stakeholder whose voice must be heard. Section 37A empowers IRDAI to prepare a scheme of amalgamation itself, without waiting for the companies to propose one, if it is satisfied that the public interest, policyholders' interests, or an insurer's sound management requires it; the insurer taking over must consent in writing. This is an active intervention power, not a passive supervisory one.

Whatever such a scheme does to the company, Section 37A(4A) fixes what it cannot do to the policyholder: he is to have the same interest in, and rights against, the insurer that emerges as he had in the one he started with. Shareholders whose rights come out lower are compensated. Policyholders are not, because theirs do not change.

Run-off is the quietest version of the same duty. An insurer closed to new business must go on servicing the policies it has already written, for as long as they last, with no fresh premium arriving to make it convenient. Beneath all this is a structural



WHEN COMPANIES CHANGE. Companies may come and go, or get merged, but policyholders' rights are unaffected

requirement: the life fund. Under Section 10, a life insurer must keep a separate life insurance fund, which stands as the security of the life policyholders, and cannot be applied for any other purpose. Section 56 carries that logic into failure. In a winding-up or an insolvency, the life assets and liabilities must be valued separately from everything else, and those assets cannot be applied to any other liability except to the extent they exceed the life liabilities. Other creditors reach them only after the policyholders have been answered.

IRDAI's July 30, 2026, amendment to its registration and amalgamation regulations goes further still: where a non-insurance holding company is amalgamated into an insurer, solvency ratio after the amalgamation must stay above the 150 per cent control level.

This is not an India-specific narrative. In the UK, a transfer of insurance business under Part VII of the Financial Services and Markets Act 2000 needs the sanction of the High Court, in place of each policyholder's consent. An Independent Expert, approved by the regulator, reports on its effect on policyholders, who are notified and may be heard. The court's concern is not

Where a non-insurance holding company is amalgamated into an insurer, solvency ratio after the amalgamation must stay above the 150 per cent control level

corporate convenience but whether any class of policyholder would be worse off.

In the US, state guaranty associations catch the policies of a member life insurer that fails: the surviving insurers are assessed to fund covered claims. The cover is not unlimited and limits vary by state, but the obligation does not vanish with the company.

INDIA IN PRACTICE

India has lived this at its most dramatic scale. The nationalising ordinance of January 19, 1956, brought in 245 entities — Indian insurers, non-Indian insurers and provident societies. They were not shut down and replaced. Their life business, with its policies and liabilities, vested in the Life Insurance Corporation when it was formed on September 1 that year. Policyholders who had bought contracts from companies whose names they knew woke up as LIC policyholders. They may not have noticed. That was precisely the point.

HDFC Life describes its October 14, 2022, merger with Exide Life as the first M&A transaction in the sector. It came in two stages: the acquisition completed on January 1, 2022, then the amalgamation, sanctioned by NCLT Mumbai and approved by IRDAI. By the company's account the whole transaction, from the September 2021 announcement, took under 14 months. The Sabka Bima Sabki Raksha (Amendment of Insurance Laws) Act, 2025, in force since February 5, 2026, extends the discipline into new territory, bringing mergers between insurers and non-insurance companies expressly within IRDAI's prior approval. It also

raises the FDI limit to 100 per cent, increasing ownership complexity and making the architecture that protects the contract within it more critical, not less.

ONE POLICY, 75 YEARS

The clearest proof either of us has seen came from one author's service in Kenya, where he processed a claim on a whole-life policy issued by the Nairobi branch of the Oriental Government Security Life Assurance Company in the late 1930s. From the records available to us, the policy ran through India's Independence in 1947 and Kenya's in 1963 without interruption. When LIC was formed in 1956, Oriental's obligations passed to its East Africa Division. In 1978, when Kenya required foreign insurers to incorporate locally, the Indian insurers there merged into Kenindia Assurance Company Ltd, born in Nairobi that December, and the portfolio passed to it. The claim was settled at Kenindia in 2013 — some 75 years after the policy was written, across the independence of two nations, one nationalisation, a change of sovereign law and three institutional transfers.

Not one of those discontinuities reduced the obligation. The contract simply continued, as contracts must be in systems built to keep them. That is what this architecture exists for: not the protection of companies, but the protection of promises.

Kumar is former Managing Director, and Sudhakar is former Executive Director, LIC. Views expressed are personal. Except where drawn from the authors' own service records, all data are from publicly available sources

What should India's long term energy priorities be?

A shift away from LPG for cooking, focus on RE storage and promotion of energy efficiency are vital, going ahead

Subodh Mathur

India's economic growth is vitally linked to using more energy. Higher output in industry, transport, agriculture, and even the production of services needs more energy. Further, economic growth increases people's demand for energy to power their appliances. Thus, there's a need to boost India's total energy supply every year, and this increasing supply must be affordable and resilient.

Further, India is committed to fighting climate change by reducing its greenhouse gas emissions, which, in turn, means reducing the use of fossil fuels. And that's another powerful reason to reduce the economy's dependence on imported oil and natural gas.

These structural features of the energy sector define India's long-run energy priorities.

First, India has become extremely dependent on LPG cooking, and must now quickly shift to other fossil fuels. This will reduce reliance on a

problematic fuel. First, LPG is largely imported, causing a stress on the country's foreign exchange. Second, LPG is subsidised for lower-income households, creating a substantial burden on government finances. And third, LPG use adds to greenhouse gas emissions.

Making households shift from LPG cooking will not be easy as it is convenient and, unlike wood or coal, it does not produce smoke. And the many households now connected to LPG are unlikely to give it up unless better alternatives are made available. Hence, the government and research institutions should prioritise the development of alternatives.

The shift away from LPG is all the more important for establishments like *dhabas* and restaurant that prepare food on a large scale.

Currently, there aren't many non-fossil-fuel alternatives to LPG. However, good progress has been made in the use of solar power for large-scale cooking.

For example, the Brahma Kumaris use solar energy to prepare thousands of



LPG COOKING. Many of the households are dependent on it

meals every day at their Mount Abu facility.

It is likely that the shift away from LPG will be led by non-profit groups. So, it is important that experts help them to make the shift.

Energy storage should be another focus area. Without cheap, plentiful storage, full use of renewable energy cannot be made. This is because renewable energy is a flow and is available only intermittently, not 24x7. At present, there are two major ways

to store renewable energy: in batteries and as green hydrogen. Batteries will surely be needed for large scale renewable electricity generation; battery-based electric vehicles are already well-known. What is perhaps less known is that hydrogen-powered forklifts and long-distance buses are already in use in India.

To make significant progress in energy storage, strong financial support and promotion from the government will be required.

Third, India needs to promote energy efficiency in homes, offices, factories, shops, and transport. The Central and State governments already have various energy efficiency schemes. But these are not getting the kind of backing being given to solar power. This is the case even though it is well known that properly run energy-efficiency schemes are more cost effective than solar power.

Hence, the burden of promoting energy efficiency would largely fall on motivated non-profit groups.

The writer is an economist with extensive public policy experience

BELOW THE LINE



Gripping uncles

Spotted at GFF 2026 was Finance Minister Nirmala Sitharaman, taking aim at India's premier league of professional party-poopers or as she wryly called them, the "Naraaz Phupha jis" (the chronically offended wedding uncles) of economic commentary. Addressing a room packed with high-octane innovators, the Minister

noted that India's youth boom isn't a distant prophecy, it's an active, high-definition reality. Yet, a certain vintage breed of commentators remains stubbornly committed to viewing a 21st-century digital surge through a 20th-century dial-up lens. "To borrow the now so famous, in social media at least, expression which the honourable Prime Minister used, our *Naraaz Phupha jis* are never short of reasons to doubt India's momentum and capabilities of our youth. Spending their time debating in television studios, whether the demographic dividend has slipped away or turned into a burden. They rely on vibe casting, feel-casting rather than actual numbers," she said. While these angry uncles are busy throwing

tantrums, Gen Z is already writing tomorrow's code, launching massive enterprises. **Confusion in the air** Aviation services are hitting a 'new high' every passing day under the watchful eyes of the Aviation Ministry. Recently, Akasa Air's Goa-Mumbai flight was to land at T2 in Mumbai, then a message was sent it will land at T1. A smart traveller told his driver to park between T1 and T2 and prayed that the aircraft lands safely. With the government deciding to shift the prevailing airport to a new Navi Mumbai location which is about 42 km from the old airport, only God can save the passengers. **Beyond the roundtable** A Chennai-based company recently

hosted what its public relations billed to the media as a 'roundtable' on its products. Invitations were sent to multiple media outlets and a few reporters made their way to the event location. As they entered the conference hall, there was, in fact, a table, which was also round, but the faces around it seemed oddly unfamiliar. It was in fact organised for an internal customer meet and greet session, with the firm's product heads holding discussions with its enterprise clients. So, a few journalists sat quietly (outside the roundtable) in complete awkwardness. **Surprise, surprise!** A minister's recent advice to officials to undertake 'surprise' visits to monitor a key scheme triggered

more surprise than perhaps intended. Stakeholders wondered why an important ministry remains without a full-time dedicated minister, with insiders claiming the minister has rarely been a regular presence there over the past two years. With staff already stretched, one village-level stakeholder suggested the minister first tap his party network to identify alleged irregularities. An officer pointed out that surprise inspections are part of the guidelines. The irony? Senior officials overseeing the scheme are reportedly reluctant to acknowledge deficiencies flagged by another ministry which is also handled by the same minister. **Our Bureaus**

ON THE CAMPAIGN
TRAIL.

The festive vibes continue on the ad circuit and some brand milestones are celebrated in unusual ways.

Pulsating ode

It's unusual to have an ad film refer back to an older commercial. Two-wheeler giant Bajaj Auto's new ad for Pulsar 150 celebrates 25 years of the motorcycle as well as the advertising legacy that helped create the sports motorcycling segment in India. Conceptualised by Ogilvy, the film is the sequel to Pulsar's iconic launch campaign of 2001. Twenty-five years later, the same two nurses encounter something familiar beneath a white sheet, triggering a sense of déjà vu. The story then unfolds with the reveal of the all-new Pulsar 150 and the derring do it is capable of. According to Suresh Nayak, Group Chief Creative Officer, Ogilvy India, "Creating the next chapter of a film that is so deeply remembered presents a very different creative challenge. The audience already knows the setup, so the opportunity was to use that familiarity to build anticipation and then surprise them with where the film goes."

Festive flip

E-commerce marketplace Flipkart has launched a rather long ad film to announce Big Billion Days 2026, its flagship festive sale event that begins on October 8. Building on last year's 'Yahan Kuch Bhi Ho Sakta Hai' campaign, the new ad, titled 'Luck Badal Sakta Hai, Yahan Kuch Bhi Ho Sakta Hai', takes the idea further. The film follows the story of an ordinary young man caught in a streak of bad luck, and how his Flipkart purchases during Big Billion Days change the course of each encounter. The film features Rohit Sharma, Sreeleela, Jackie Shroff, Aneet Padma and Samay Raina, who appear at different points in the protagonist's reset day. The campaign has been conceptualised by Leo Burnett.

Vi for all

Following the recent launch of its refreshed brand identity, telecom operator Vi has released a new ad film, 'Vi pe sab hain VIP', starring Shah Rukh Khan. Using humour and Shah Rukh Khan's larger-than-life persona, it dramatises its promise of delivering improved experiences, services and benefits for all its customers. In the film, Shah Rukh Khan assumes that Vi's latest upgrades and offerings have been launched especially for him, until he is made to realise that they are, in fact, for everyone on Vi, leaving him surprised. The film is built around the core philosophy of "We Love All". The campaign has been conceptualised by Ogilvy and brought to life by Chrome Films.

Sweet comfort

The latest Hershey's India ad campaign, 'Life Is Hard, Par Hershey's Hai Na Yaar', is based on the idea that when the world feels like a lot, its signature chocolate can offer comfort. The film follows a protagonist through a chaotic day that gets hilariously worse. Just when it feels like everything is lost, Hershey's steps in with a sweet moment of joy. Created by TBWA\Lintas, there is an innocent optimism that pervades the film.

● PLAYING WITH FIRE

Onida rewired: The devil is in the details

PRODUCT TRUTH. Onida bets on its iconic 'devil' as it attempts a fifth comeback. But can nostalgia alone tempt the younger buyers?

Amit Vijay Mohile

“A known devil is better than an unknown one” is the common saying. For Onida, familiarity may no longer be enough. Its ‘devil’ may need to take an unfamiliar turn as a crusader taking on problems that appliance buyers have learnt to tolerate — from hard water damaging washing machines to air conditioners that force households to buy separate heaters during winter, and obscure warranties and repeat repairs. That would give a contemporary cause to a mascot introduced in 1984 with the tagline “Neighbour’s Envy, Owner’s Pride”, but repeatedly retired, revived and reintroduced since. Harish Bijoor, brand strategy specialist and founder of Harish Bijoor Consults Inc, argues that Onida’s devil needs precisely such a role for a new generation. “Make the devil the Tukaram Mundhe of consumer durables activism,” Bijoor says, invoking the bureaucrat known for taking on entrenched systems. His argument: Don’t make the devil Onida’s salesman; make him the consumer’s advocate. Sanjay Sarma, Principal Consultant and Founder of SSARMA Consults, agrees these are the “newer problems” that Onida could own and almost “fix the plumbing for your life”. ORIGINS IN ENVY Giraj Sharma, Co-founder of BehindTheMoon Consultants, who joined Onida in 1985 and spent 12 years heading marketing, recalls that colour TVs were scarce enough for families to gather at neighbours’ homes that had them. The envy ex-



COME HITHER. Will Gen Z know, or care, for Onida’s yesteryear anti-hero

isted before the advertising did; Onida’s tagline captured that tension, and the devil gave it a face. Onida’s brand manual now moves him “From Envy to Intelligence”, replacing “I want what they have” with “Maybe there is a smarter choice”. The stakes are bigger than just another advertising comeback. At its peak in the late 1980s and early 1990s, Onida commanded 18–20 per cent of India’s colour TV market. Revenue fell about 45 per cent from ₹1,192.17 crore in FY22 to roughly ₹660 crore in FY26. Under Onida 5.0, its devil is recast as an “intelligent provocateur”, darker, sharper and more premium — one who “never follows”, “never attacks” and “never boasts without reason”. His confidence, the manual says, comes from “product truth”. RENAMING AND REBOOTING But while Onida has defined how its resurrected devil should behave, it’s less clear which consumer problem he is out to solve. For “Onida rewired” the company has raised ₹149.52 crore, exited contract manufacturing and renamed MIRC Electronics as

When brands come back to the fold

From umbrellas to mobile phones, unfolding the benefits of foldable products



Earlier this month, Apple launched its first foldable iPhone, the Duo. The starting price in India is nearly ₹3 lakh. Advanced versions are priced more than ₹4 lakh. The two questions on many minds are: “Why does anybody need a foldable phone?” and “Why would consumers pay such a huge price for this device?” Foldable phones have generally not been very popular. But I do know people who use them with passion and pride and show them around. For instance, someone who uses another foldable phone brand, the Samsung Galaxy X Fold8, recently told me how the large screen enables him to work seamlessly while travelling. He also took pains to explain some of the intricacies of the folding screen. It is, therefore, likely that the iPhone Duo, too, may soon have its band of avid users in India. Many

may feel it enhances their productivity and ease of use. Others may see it as a marker of status. The world may soon be split between the “Duolites” and the rest of us.



COMPACT STATUS. The Duo is the iPhone’s first foldable act BLOOMBERG

UMBRELLAS AND LAPTOPS But if this makes you think foldable products are elitist, you are mistaken. The original foldable product is the umbrella — a common man’s product. Unless an umbrella folds, how can you carry it around or store it conveniently? In fact, there is even the four-fold umbrella, which collapses into four sections, making it ultra-compact. It can easily fit into a small purse or a trouser pocket. So you can carry an umbrella in your pocket and display a foldable phone proudly in your hand, which is the reverse of what many of us currently do. Another product where we take the fold for granted is the laptop or notebook computer. Desktop computers were not foldable, and the first few notebook computers, too, did not fold. The clamshell design in use today, where the screen rests flat on the keyboard once folded, was first introduced

by a brand named Grid Compass in 1982. It was initially expensive, and primarily used by NASA during early space shuttle missions. Today, it is the default design, enabling us to carry our laptops with ease. CHAIRS AND SWISS KNIVES Then there is folding furniture, including chairs and tables, which have been around for decades. A bed that folds into a compact sofa has saved space in many homes. I own a Swiss knife and this in-

novative product always amazes me — multiple knives, scissors and other tools fold into its beautiful, lacquered red handle, and sit on top of each other in a stack, which forms the heart of its design. The Swiss knife is a great multitasker and a true champion of folded products. From Japan comes the art of origami — folding paper to create interesting and beautiful designs. Having nothing to do with convenience, the folded products here offer us art and beauty in our daily lives.

And let me not forget books, where the pages fold so elegantly over a spine, enabling us to open and close the book smoothly.

THE BENEFITS The range of folding products bring home the functional benefits of a fold. Sheer convenience in the case of the umbrella or laptop computer, versatility in the Swiss knife, and a wider screen in the case of a foldable mobile phone. There are also the emotive benefits. A four-fold umbrella in your pocket is quiet reassurance that you will not get wet. A folding bed makes us feel good about being a gracious host when guests stay over. Origami makes us feel beautiful. And, to return to where we began, a small foldable phone can be a big marker of status. So, will the iPhone Duo be a big hit in India? My view is yes, because both the functional and emotional benefits of this device are irresistible for a clear segment of our population. The actual answer will, of course, unfold over the next few months.

Harish Bhat is a curious marketer and bestselling author

● PERFECT MATCH

India-Brazil football love story: Samba, spice and extra tickets

Anirban Som

What if Brazil and India married? Not literally, of course. But how about a brand “match making” — organising a fun shaadi — ahead of the India-Brazil football fixture in Kolkata on October 3? Influencers could represent the two cultures, food creators could blend cuisines, fashion creators could design match-day outfits, and musicians could remix each country’s football chants. Comedians could play the fictional families, with chaacha-bua dispensing unsolicited advice and creating drama over the guest list. The brand’s social media handles could become the epicentre of the wedding drama. Brazil demands Samba at the sangeet, while India wants Bollywood. Followers vote on whether pao de queijo (Brazilian cheese roll) or pani puri belongs on the menu. Then comes a mock crisis: Both sides want extra tickets for the match and the elders intervene.

The idea may sound outlandish at first, but that is precisely why it could work. India’s three-match run involves Panama in Bengaluru on September 25, Brazil on October 3 and Uruguay on October 6 in Kolkata. Brazil are five-time World Cup champions, Uruguay are two-time champions, and Panama featured at the 2026 World Cup. For brands, however, the opportunity could go well beyond putting logos around the stadium or running match-day advertising. SHAADI AND STORIES A brand could create micro-films on the many ways in which Brazilian football has shaped India’s game. They can narrate stories about legendary stars such as Jose Ramirez Barreto, a Mohun Bagan favourite; Cristiano Junior, whose goals helped East Bengal win the 2003-04 National Football League; Douglas Silva, who represented both East Bengal and Mohun Bagan in the early 2000s; and the current crop such as Brazilian Miguel Figueira, who



FAMOUS FEET. Kolkata splashes its legendary love for Latino football

was part of East Bengal’s 2025-26 ISL campaign before moving to Mohun Bagan. Then there is Pele’s 1977 visit — perhaps the most powerful story connecting Brazil and Kolkata. The football great arrived with the New York Cosmos for a much remembered match against Mohun Bagan at the Eden Gardens, watched by around 80,000 fans. Imagine that story told through the eyes of a grandfather who was among the fans who packed the stadium that day, reliving the thrill of the moment for his young grandchild. The films could be showcased on short-form content platforms or apps such as Pocket Films and Kuku TV. The sponsor need not dominate the stories, but merely enable them. The opportunity is significant. The Google-Deloitte 2024 Think Sports report estimated that In-

dia has 85 million football fans, and 90 per cent of its sports fans follow more than one game. Entertainment can provide the entry point: Curiosity leads to attention, engagement and fandom. BEYOND THE MATCH The three matches could be ripe with opportunities beyond marketing. Imagine an Indian youngster catching the eye of a Brazilian academy, an Indian youth team training abroad or local coaches participating in sessions with Brazilian support staff. For young players, simply being around elite professionals such as Vinicius Jr (Brazil) and Federico Valverde (Uruguay) could offer lessons no training manual can replicate — from movement and finishing to handling pressure. Sports psychologists could share approaches to player development and mental preparation, media professionals could exchange ideas on building the digital presence of clubs and players, and coaches could be motivated to pursue higher-level coaching licences and qualifications.

Uruguay already offers a compelling example. In 2024, Indian footballer Bijay Chhetri joined Uruguay’s Colon Futbol Club on loan, later signing a permanent contract — the first such move by an Indian footballer to a Latin American club. Brazilian expertise is already in use to upgrade the playing turf at Kolkata’s Vivekananda Yuba Bharati Krirangan, or Salt Lake Stadium. FANS AND FOLLOWERS Kolkata, in particular, has a rich football legacy — from the storied East Bengal-Mohun Bagan rivalry to iconic players and memorable matches. There is no lack of stories. What’s missing are the re-tellings for a new generation of fans. Look at Manchester United. Its history, tragedies, triumphs and legendary figures are constantly woven into the fan experience through digital content, fan experiences and museum spaces. The Munich air disaster of 1958, the loss of eight members of the famous Busby Babes, the rebuilding that followed and the young

players who stepped up to carry the club forward and reach the FA Cup final just three months later — these are not merely historical facts but also a part of the club’s living identity. Could India take a cue and bring its history of the game closer to fans through a temporary football museum or exhibition near the stadium, or by promoting existing spaces such as the East Bengal FC Museum in Kolkata? That should ultimately be the measure of these friendships. Not just how many logos were seen or how full the stadium was, but also more people following Indian players, greater interest in clubs and academies, stronger international connections and a larger audience returning to Indian football. Brazil, Uruguay and Panama can be catalysts for change in Indian football, bringing the world to India. The bigger challenge — and the bigger prize — is to make Indian football worth coming back to.

Anirban Som is Associate Professor, Marketing, at IIM-Tiruchirappalli

STURDY COURIER



BonV Aero develops heavy-lift UAV for high-altitude regions

M Ramesh

Bhubaneswar-based drone maker BonV Aero, which is one of the five finalists selected for the GENIUS NY accelerator programme of the State of New York, is developing an ultra heavy payload, high-altitude drone capable of carrying more than 100 kg to altitudes above 10,000 ft and over distances exceeding 100 km.

The capability is particularly relevant to military logistics in India's mountainous and other high-altitude regions. BonV Aero is already a supplier of logistics drones to the Indian defence sector.

Its existing drone can carry a 50-kg payload, fly above 10,000 ft and remain airborne for about 30 minutes, the company's Co-founders and CEOs Satyabrata Satapathy and Gaurav Achha told *businessline*.

Their new drone is designed to tackle challenges specifically involved in operating at high altitudes, where the thinner atmosphere makes it more difficult to generate lift. According to Satapathy, a key differentiator in BonV Aero's drones is their ability to operate in such conditions.

Unlike the carbon-fibre structures commonly used in drones, BonV Aero's aircraft use an aluminium alloy monocoque body. Satapathy said the choice was made taking into account the very low temperatures encountered at high altitudes.

Another feature, according to the company, is the use of multiple rotors that continue to contribute thrust during the cruise phase, rather than becoming redundant after take-off. The forward and rear rotors are operated at different speeds, with the rear rotors being faster, while the propulsion system combines batteries with a fuel-cell system.

BonV Aero says its drones are designed and manufactured entirely in-house, with no imported components. The drones are light enough that just two people are needed to carry them and deploy them anywhere. Satapathy said this enables the company to offer its drones at about one-fifth the cost of comparable Chinese products.

The company is currently building a manufacturing facility in Bhubaneswar at an investment of about ₹300 crore. The plant's planned production capacity is 500 drones a month, compared with the current capacity of about 200 drones a month.

The selection for GENIUS NY gives BonV Aero entry into a year-long, in-residence programme with product and business development support, access to investors and industry leaders, and exposure to the growing uncrewed systems ecosystem in New York.

The five finalists will compete for a total of \$3 million in direct investment — a \$1 million grand prize and four awards of \$500,000 each.

For BonV Aero, the programme also offers a route to the US market for its heavy-lift unmanned aircraft.

GRID STABILITY

HOME ADVANTAGE. How local minerals can help solve India's solar curtailment problem at a lower cost vis-a-vis Li-ion batteries

K Bharat Kumar

India's accelerated efforts to raise renewable energy capacity in the coming years face at least one significant challenge — lack of mature storage technologies. Renewable power generation firms are often forced to stop production at midday, which is peak hour for solar power production, as there isn't enough demand for such energy.

Theoretically, power can be stored in largescale batteries and used when demand rises in the evenings. But the country does not have such storage capacity due to the high cost of imports and lack of indigenous technology.

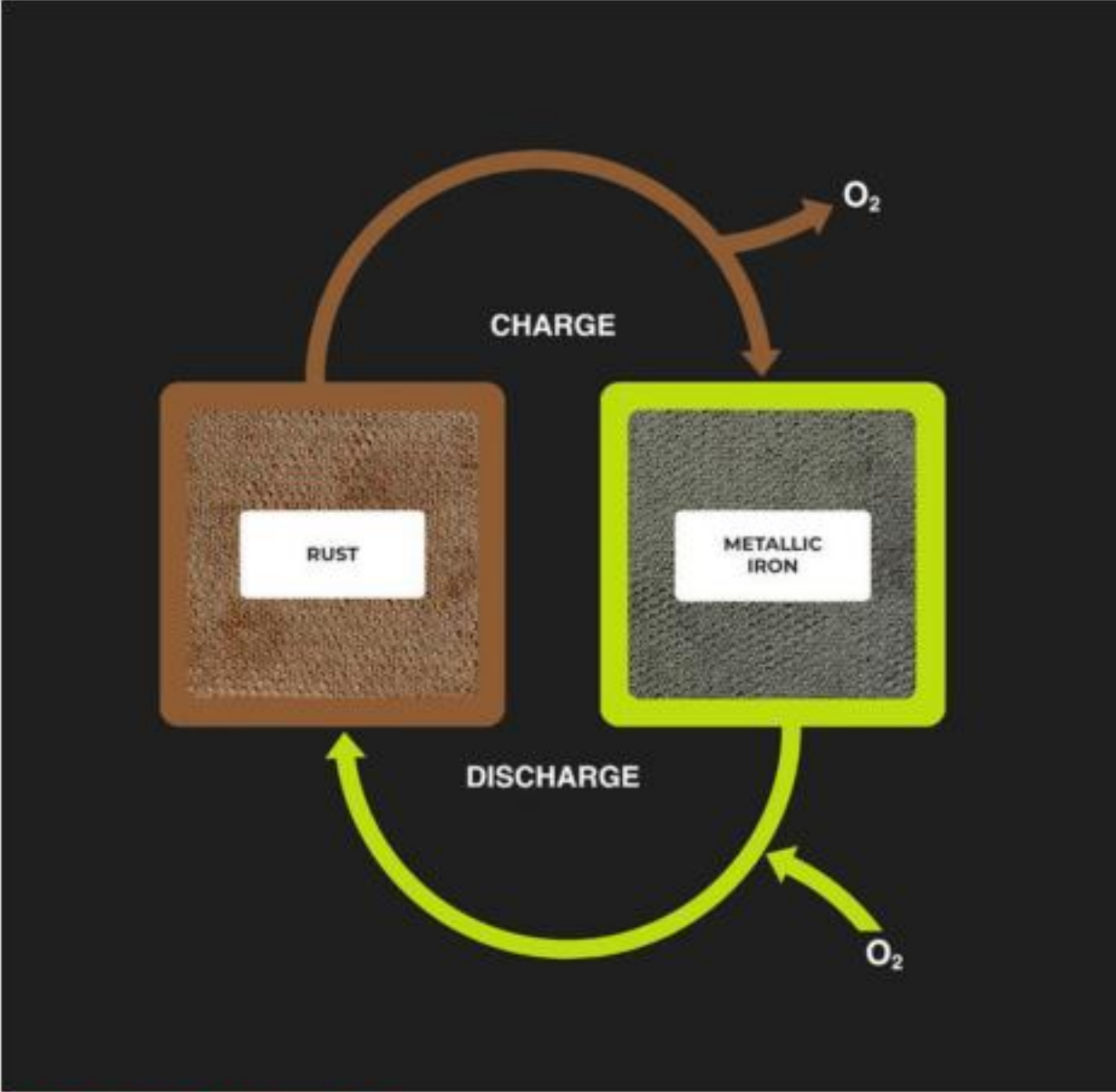
Producers have seen 15–20 per cent average curtailment — namely having to stop production due to lack of demand — with at least one instance of curtailment spiking to 65 per cent last year due to lack of grid-scale storage.

According to Prof Aravind Kumar Chandiran, Faculty, Department of Chemical Engineering, IIT-Madras, "We see from our studies that electricity consumption is projected to triple in India from about 2,000 terawatt hour (TWh) today to about 6,000 TWh by 2047."

He reasons this would mean about 4 TWh of daily energy storage capacity would be needed, assuming only 50 per cent of power would come from renewables and only half of this would need storage for future use.

As India cannot afford to rely only on imported lithium to meet storage demand of such magnitude, it is important to bring in alternatives, such as domestic mineral solutions.

Lithium-ion batteries offer high power delivery and are quickly responsive. They are ideal for electric mobility and transient, high-power



PROVING METTLE. Iron-air battery workflow; and Priyansh Mohan, Co-founder and CEO of Meine Electric, with the firm's fast-charging iron-air battery system

grid spikes. Metal-air technologies (such as iron-air or zinc-air) offer low power delivery but for a longer duration. These are best suited for steady-state grid storage.

Unlike lithium-ion battery manufacturing, which requires expensive, highly controlled clean-room and ultra-dry environments to prevent chemical instability, iron-air systems can be fabricated in ambient atmospheric conditions, drastically reducing production and capital expenses.

Metal-air systems suffer from what is called 'high electrochemical overpotential' — they need more voltage during a reaction, such as the charging or discharging of a battery. This results in significantly lower efficiency compared to lithium-ion technology. For example, iron-air yields an efficiency of 23–50 per cent, while for zinc-air it is 40–80 per cent, according to Prof Chandiran.

Notably, he mentors this field directly, with three of his former PhD students having founded a

start-up focused on commercialising indigenous zinc-air battery technology for stationary grid storage. Further, lithium-ion battery is a mature technology, whereas metal-air batteries still face chemistry hurdles, engineering scaling challenges and a lack of long-term real-world testing.

LONG-DURATION STORAGE

Startup Meine Electric disclosed in July that it had developed the world's first fast-charging iron-air battery system. This could likely offer a breakthrough in long-duration energy storage.

Iron-air batteries generally operate on a basic reduction-oxidation mechanism — iron reacts with oxygen and water during discharge to form iron oxide, releasing electrical energy. Reversing this reaction forces the iron oxide back into metallic iron.

"Traditional iron-air systems are restricted to multi-day cycles, making them largely incompatible with the intermittent nature of renewable

energy generation," says Priyansh Mohan, Co-founder and CEO of Meine Electric. "Solar-heavy grids don't need a 100-hour battery. Unlike regions that require seasonal storage, the renewable challenge across Asia is fundamentally a daily balancing problem." He says Meine has engineered its system to capture a full charge within a six-to-eight-hour window. This has helped compress the multi-day charging window to six hours, paired with 18 hours of discharging.

The company says its technology has been tested and validated by the Customized Energy Solutions (CES) Battery Laboratory. The lab confirmed the "operational stability, capacity retention and electrochemical performance under this 6/18-hour cycle", Meine stated.

According to Mohan, using Meine's proprietary technology, the storage cost over the battery's lifetime would be about ₹5 per kWh. The clincher here is as much the low cost of operation as the raw material iron, as India has an abundant

energy generation," says Priyansh Mohan, Co-founder and CEO of Meine Electric. "Solar-heavy grids don't need a 100-hour battery. Unlike regions that require seasonal storage, the renewable challenge across Asia is fundamentally a daily balancing problem." He says Meine has engineered its system to capture a full charge within a six-to-eight-hour window. This has helped compress the multi-day charging window to six hours, paired with 18 hours of discharging.

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We value your feedback.
Do send your comments to
quantum@thehindu.co.in

WOW. In search of a mate



M Ramesh

In the summer of 2025, three fellows who, by all accounts, were desperate left their home in Kafue, Zambia, to look for females to mate with. Their world was shrinking — not many opportunities. Little did they know they were marked. Somebody kept an eye on them. They travelled on foot far into wild, unknown territory. When they encountered cities they skirted them. They came to rivers and walked along the bank until they found an easy point to cross. They walked around large reservoirs. They persevered doggedly.

In April 2026, they reached a place that was 420 km from where they started, and they had walked 2,506 km to get there. And then they walked no more. They appeared to have "settled".

Their watchers surmise that they found their females.

After all, the Luano Game Management Area in the Luangwa National Park is known to hold a large population of African wild dogs. This remarkable journey has been recorded in the recent issue of *Ecology*, in a paper titled 'Extraordinary movements connecting distant ecosystems by African wild dogs'. Such journeys have been recorded before — animals go a long way to find mates. The *Smithsonian* magazine, which has a story from the paper, highlights a telling message from the dog episode: The importance of connected landscapes for conservation.

Today, with tagging and GPS, it is possible to take a close peek into the fascinating lives of animals. Jack London could only imagine it in his 1903 adventure novel *The Call of the Wild*. London took his canine hero, Buck, from his cosy home in California to the wilderness of Alaska, where he joined the other descendants of his ancestors — thus truly getting back home.

HEATING UP

Headway in solar thermal

M Ramesh

The Ministry of New and Renewable Energy has given a 'call for proposals for concentrating solar thermal technology'. Solar thermal — heat rather than electricity from sunlight — is an efficient use of solar energy. Against the backdrop of the MNRE's call, it is useful to look at two solar thermal technologies cooking in the labs.

One, according to details published in *Energy*, is about combining low-temperature solar thermal collectors and high-temperature heat pumps, using ethanol as the working fluid (refrigerant).

Industries often need heat at 120–200 degrees C. Solar thermal systems typically deliver much less. However, by combining with heat pumps, researchers at the University of Guanajuato and Texas A&M achieved 155.6 degrees C.

The process is simple. Step 1 is to use solar collectors to heat water to about 99 degrees C. The hot water is stored in an insulated tank. Step 2 is to use the hot water to heat the pump's evaporator, turning the refrigerant (ethanol) into vapour. The third step is to raise the refrigerant's temperature by compressing the vapour.

In the industrial process under study, the refrigerant reached about 156 degrees C — enough heat for making tomato paste — before condensing and returning through an expansion valve for the next cycle.

The heat pump effectively acts as a thermal elevator: Solar energy enters at a low temperature, and electricity drives the compressor to deliver useful heat at a considerably higher temperature.

The researchers tested four working fluids. Ethanol emerged the strongest performer. At a temperature lift of 70 degrees C, the heat pump achieved a coefficient of performance (COP) of 4.64, which means that every unit of compressor electricity was associated with 4.64 units of heat delivered. The calculated 'exergy efficiency' was 74.4 per cent, while the levelised cost of energy was \$0.0818 per kWh.

The result is significant because the solar collectors themselves do not have to operate at the process temperature. Instead, relatively inexpensive low-temperature solar heat is upgraded electrically.

The authors estimate that the system could avoid about 3,613 tonnes of carbon dioxide a year, and the ethanol-based configuration had a simple payback period of about 8.5 years in the case studied.

The second study, published in the *Journal of Energy Storage*, is about making an age-old idea — compressed air storage — work.



BOOSTER. A heat pump can act as a thermal elevator for solar collectors ISTOCK

You can store energy by compressing air using electricity when available cheaply. But the problem arises during the discharge cycle, when the air is let out to drive turbines. When let out, the compressed air decompresses and becomes cold, leading to problems such as ice formation on turbine blades. However, more importantly, cold air lacks energy and cannot turn turbines.

A group of researchers from Spain, Italy and Switzerland have tackled this problem with a ridiculously simple trick: Use solar thermal energy to re-heat the compressed air before it enters the turbines. The researchers modelled this in Chile. They used electricity to compress air into an underground cavern, while solar energy heats air and this goes into a thermal energy storage system. During discharge, the stored heat is used to reheat the compressed air before it expands through the turbines.

There is another useful twist. The heat released during the compression can be recovered and supplied to a nearby industrial user, thereby improving the economics of the system. The researchers envisage industrial heat of 140–160 degrees C.

The economic viability of the two systems depends on factors such as the cost of input electricity and the prices that the output electricity and recovered heat will fetch. But they clearly show a pathway that can be improved.

Neither technology is yet a universal competitor to conventional systems. The 'compressed air energy storage with concentrated solar power' (CAES-CSP) study, for instance, puts the concept only at technology readiness level (TRL) 3–4, compared with TRL 9 for battery energy storage systems and stresses that its economics depend on the unusual coincidence of strong solar resources, suitable geology and a nearby industrial heat consumer.

But that may be precisely where the opportunity lies: Instead of asking how much electricity a solar plant can produce, ask what combination of electricity, heat and storage it can deliver.

When solar glass becomes a power generator

US company Caelux achieves this by coating the glass with a thin film of perovskites

M Ramesh

Solar cell and module manufacturer GREW Solar recently announced a five-year commercialisation partnership with US-headquartered Caelux Corporation to produce high-power hybrid-tandem solar modules using Caelux's 'energy producing glass'. Here, the glass covering the solar cells itself is an energy producing layer.

Caelux achieves this by coating the glass with a thin film of perovskites. Vish Iyer, Managing Director-India and West Asia, says the perovskite layer captures sunlight that the silicon cell cannot use as efficiently, while the cell itself continues to perform as before.

The solar industry has focused on two approaches to perovskite technology: Integrating perovskite into the silicon solar cell and developing perovskite as an energy pro-

ducing layer. Caelux's R&D focused on the latter, Iyer tells *businessline*.

The distinction is important. In the conventional perovskite-silicon tandem approach, the perovskite is deposited directly on the silicon cell, requiring changes to cell architecture and perhaps new manufacturing equipment. Caelux, instead, deposits the perovskite solar layer on the glass that normally serves as the module's front cover.

"In effect, we turn that inert glass into an energy producing layer," Iyer says.

This allows the use of existing silicon module factories and equipment. Caelux says its energy producing glass can be paired with mono-PERC, TOPCon, back contact and heterojunction technologies, and with modules 1.7–2.5 m long.

Rather than replacing silicon, it is about adding another generator to the same module.



SURFACE POWER. Caelux solar modules with 'energy producing glass'; and Vish Iyer, Managing Director-India and West Asia, Caelux



For instance, a 10 GW TOPCon manufacturing line could, according to Caelux, produce the equivalent of 12 GW of module output after incorporating its technology with limited additional investment.

The higher power density could also reduce the requirement of land, mounting structures, cabling and other components.

Caelux has signed agreements with Indian manufacturers, including Rayzon Solar and Navitas Solar, for 5

GW hybrid-tandem production. Together, with the GREW Solar deal, its contracted volume with module manufacturers globally is 19 GW.

The Indian partnerships target commercial production from 2028. Navitas and Rayzon target 28 per cent efficiency with TOPCon-based modules, while GREW's modules target efficiencies above 29 per cent.

"Perovskite-silicon tandem could be an important breakthrough for the solar industry, enabling significantly higher power from the same module

GLOBAL BYTES.

flydubai inks pact with Air India

flydubai has announced new interline agreements with Air India and Uzbekistan Airways, expanding its global network across Asia, Australia and Europe.

These partnerships offer flydubai passengers seamless access to more than 80 additional destinations through partner networks, while offering customers of Air India and Uzbekistan Airways convenient connections to key destinations across the airline's network, via Dubai, with single-ticket itineraries and through-checked baggage.

With these additions, flydubai's interline network now comprises 46 partner airlines, reinforcing the carrier's strategic role in enhancing Dubai's connectivity and strengthening the emirate's position as a global hub for travel and commerce, a media release stated.

Panama Canal bulkier transits drop 22%

"Since the start of July, Panama Canal bulkier transits have fallen 22 per cent year-on-year. Bulkier have faced rising competition for transit slots from other sectors, especially tankers, as the US has boosted energy exports to Asia. El Nino is also expected to lead to lower water levels, which means transit restrictions and heightened competition for available capacity," says Filipe Gouveia, Shipping Analysis Manager at BIMCO.

The canal is commonly used on trade routes between the east and west coasts of the Americas, as well as some routes between the Americas and East Asia or Europe. It is important for the US, which is the origin or destination of 54 per cent of the dry bulk cargoes transported through the waterway. Grains, steel, fertilizers, coal, salt and petcoke are the main dry bulk commodities transported via the canal, primarily by supramax, handysize and panamax ships.

"The decline in bulkier Panama Canal transits has resulted in rerouting via the Cape of Good Hope or Cape Horn. The diversions increase average sailing distances and boost tonne-mile demand. During the previous El Niño in 2023-24, when bulkier transits decreased significantly, rerouting increased global dry bulk tonne-mile demand by around 1 per cent," says Gouveia.

Waivers under the PARIVARTAN scheme to promote clean commercial transport

- No road tax**
100% road tax waiver for eligible new bus and truck
- No registration fee**
100% registration fee waiver for trucks and buses
- Savings on upgrade**
Reduced upfront cost of moving to a cleaner modern bus or truck

Source: Ministry of Road Transport & Highways (MoRTH)

RE POSTS.

- NW-2 is set to facilitate methanol transport for Assam Petro-Chemicals Ltd. to Bangladesh!
A vessel with 27 ISO tanks is sailing from Guwahati to Bogibee to load methanol, the newest commodity to be transported through inlandwaterways in the North Eastern Region.
IWAI
@IWALShipMin
- The Salt Container Train marks a new milestone in freight connectivity, enabling efficient movement of salt consignments in specially designed containers strengthening salt transportation.
Ministry of Railways
@RailMinIndia
- On the cargo apron at ChennaiAirport, every movement follows a sequence.
From bringing shipments to the aircraft stand to loading them into the aircraft, ground handling teams coordinate each step as part of the turnaround.
A closer look at the everyday work behind an air cargo movement.
Chennai (MAA) Airport
@aachnaiport

Truck rental rates

Truck route	Rentals as on		% change
	Sep 1	Sep 17	
Delhi-Mumbai-Delhi	1,75,000	1,76,000	0.57
Delhi-Nagpur-Delhi	1,63,000	1,62,000	-0.62
Delhi-Kolkata-Delhi	1,82,000	1,81,000	-0.55
Delhi-Guwahati-Delhi	1,96,000	1,97,000	0.51
Delhi-Hyderabad-Delhi	2,05,000	2,06,000	0.49
Delhi-Chennai-Delhi	2,20,000	2,21,000	0.45
Delhi-Bengaluru-Delhi	2,29,000	2,30,000	0.43
Delhi-Ranchi-Delhi	1,75,000	1,76,000	0.57
Delhi-Raipur-Delhi	1,68,000	1,70,000	1.18
Delhi-Kandla-Delhi	1,46,000	1,45,000	-0.69
Mumbai-Chennai-Mumbai	1,75,000	1,76,000	0.57
Ludhiana-Hyderabad-Ludhiana	2,05,000	2,06,000	0.49
Mumbai-Kolkata-Mumbai	2,07,000	2,06,000	-0.49
Chennai-Ahmedabad-Chennai	2,16,000	2,15,000	-0.47
Bengaluru-Kolkata-Bengaluru	2,18,000	2,20,000	0.91
Bengaluru-Mumbai-Bengaluru	1,50,000	1,51,000	0.66
Delhi-Thiruvananthapuram-Delhi	2,88,000	2,86,000	-0.70
Guwahati-Mumbai-Guwahati	2,87,000	2,88,000	0.35
Nagpur-Chennai-Nagpur	1,71,000	1,72,000	0.58
Kolkata-Guwahati-Kolkata	1,63,000	1,62,000	-0.62
Indore-Delhi-Indore	1,52,000	1,51,000	-0.66
Ahmedabad-Indore-Ahmedabad	1,05,000	1,06,000	0.94
Vijayawada-Mumbai-Vijayawada	1,68,000	1,69,000	0.59
Vijayawada-Kolkata-Vijayawada	1,57,000	1,58,000	0.63
Mumbai-Nagpur-Mumbai	1,20,000	1,21,000	0.83
Mumbai-Pune-Mumbai	75,000	76,000	1.32
Mumbai-Aurangabad-Mumbai	88,000	87,000	-1.15
Mumbai-Nashik-Mumbai	74,000	75,000	1.33

Figures in ₹ Source: Subham Freight Carriers India Pvt Ltd

EXPENSIVE CELEBRATION

Fewer flight seats for festival travel

PRICE OF JOY. Domestic air tickets turn costlier and elusive as airlines hold back capacity additions during busy season

Rohit Vaid

It's the time of the year when crores of Indians pack their bags for get-togethers with loved ones to celebrate a string of festivals or head to a tourist destination for rest and recreation. At the start of the season, which extends from September to January, domestic air ticket prices are ruling above last year's levels, even as the fares have eased from August, offering travellers partial relief.

Data shared by Cleartrip showed that average domestic airfares for September are 12-20 per cent higher compared to a year ago, while they have declined by 5-10 per cent compared with August.

ixigo's year-on-year comparison showed that average one-way fare on the Delhi-Hyderabad sector rose 74 per cent to ₹8,253 from ₹4,739. The Delhi-Goa fare increased 47 per cent to ₹7,896 from ₹5,377, while the Delhi-Bengaluru fare rose similarly to ₹9,199 from ₹6,273. The Mumbai-Kolkata fare increased 43 per cent to ₹8,094 from ₹5,653.

The rates are for bookings made 15-20 days before departure.

The comparison also showed that the Delhi-Pune fare rose 35 per cent to ₹7,150 from ₹5,293, while the Delhi-Mumbai fare increased 21 per cent to ₹6,456 from ₹5,317. The Mumbai-Bengaluru fare rose 45 per cent to ₹5,884 from ₹4,066.

SEAT CAPACITY

The higher fares come at a time of fewer scheduled seats in the domestic market.

According to consolidated industry data from aviation analytics firm Cirium, scheduled seat capacity for September was down 5.3 per cent year-on-year to 1.49 crore from 1.58 crore.

This implies 8.32 lakh fewer scheduled seats compared with September last year. The number of scheduled flights is down by 8.5 per cent to 83,515 from 91,287 —



ELEVATING TREND. Average domestic airfares for September are 12-20 per cent higher compared to a year ago BLOOMBERG

namely 7,772 truncated services. Meanwhile, capacity measured in available seat kilometres (ASKs) is lower by 6.2 per cent. This measure captures both the number of seats offered and the distances of scheduled services.

The Cirium figures describe scheduled capacity for September, rather than flights actually operated or passengers carried.

Sequentially, the number of scheduled domestic flights for September are 4 per cent lower compared with 87,035 in August. Seat capacity is also down 4 per cent from 1.55 crore, while ASKs are lower by 4.7 per cent.

Further, the scheduled seat capacity in September is fewer by about 6.29 lakh than in August. These industry totals do not establish whether capacity has fallen on all routes where fares have risen.

For those planning to travel during the festival season, capacity additions will be a key factor shaping ticket prices, an industry insider pointed out, adding that fares through the year-end would depend on the extent of additions by airlines.

JET FUEL PRICE RISE

Other industry observers said elevated jet fuel prices linked to the West Asia crisis and lower domestic capacity contributed to the annual rise in airfares.

However, a senior executive with a leading airline tells *businessline* that the increase in airfares would not be sufficient to offset the sector's operating losses.

Another senior aviation executive says the weakening rupee against the dollar is adding to the cost pressure. The executive estimates that around 70 per cent of

airline costs, including aviation turbine fuel (ATF), is denominated in, or linked to the dollar.

In addition, the fare increases would need to cover both higher ATF costs and the currency depreciation to enable airlines to maintain bottom lines at last year's level, the executive says.

Cleartrip's monthly progression showed June had the highest fares in the June-September period, reflecting peak summer pricing.

Fares declined 12-18 per cent in July, followed by a 5-8 per cent increase in August, before easing again in September.

We value your feedback.
Do send your comments to
bl.logistics@thehindu.co.in

FRESH TRACKS

Trade, industry hail Mangaluru rail rejig



MONEY SPINNER. The railway marshalling yard at the New Mangalore Port

A J Vinayak

Stakeholders are hopeful that the Railway Board's decision to merge the Mangaluru railway region with the Mysuru division of South Western Railway (SWR), effective October 1, will boost the prospects of trade and industry in coastal Karnataka.

At present, the railway network in the Mangaluru region is shared between the Palakkad division of Southern Railway (SR), Mysuru division of SWR and Konkan Railway Corporation Ltd. A 30-km stretch in the Mangaluru region, currently under SR, will be merged with SWR on October 1.

Welcoming the change, PB Ahmed Mudassar, President of the Kanara Chamber of Commerce and Industry (KCCI), says: "Our railway network has been split across three zones for too long, with coordination gaps that delayed new trains, service upgrades and infrastructure spending. This has been a shared cause for passenger groups, civic organisations and industry alike."

He says the decision unlocks new opportunities, while integrating the region with the rest of the State. It will enable local businesses to access newer markets, create greater opportunities for local goods and have a multiplier effect on the regional economy.

FREIGHT REVENUE

In one of the memorandums to the Railway ministry, former KCCI President Ananthesh Prabhu had stated that the Mangaluru region accounts for 90 per cent of the freight revenue of the Palakkad division. "Be it better infrastructure, maintenance, new rail service and many more needs, the Mangaluru station has always been dependent on the mercy of Palakkad division and grossly neglected," he said.

Agreeing with this, N Kumar, an expert on the region's logistics sector, says

the revenue generated at the railway marshalling yard at the New Mangalore Port (NMP) is currently apportioned to the Palakkad division. Despite its revenue generation, Mangaluru has not received much attention towards freight infrastructure development, he says.

The Palakkad division also controls the allotment of rakes for freight handling.

Kumar hopes that the upcoming merger would improve both railway traffic and railway infrastructure development in the Mangaluru region. Availability of rakes could improve, and the revenue generated in the Mangaluru region could be better utilised for enhancement of facilities, he says.

JURISDICTIONAL HURDLES

Capt Brijesh Chowta, Member of Parliament from Dakshina Kannada, who pursued the case to bring Mangaluru region under Mysuru division with the authorities concerned, says the Railway Board's move reduces jurisdictional hurdles, improves coordination, enables faster resolution of passenger concerns and gives greater attention to railway infrastructure and connectivity across coastal Karnataka.

GG Mohandas Prabhu, former member of the Palakkad division railway users' consultative committee, tells *businessline* that the region's trade and industry will have to come out with a blueprint for the improvement of traffic and infrastructure development in the Mangaluru region. "We have to highlight the revenue generated from the region and seek proportionate facilities."

A plan should be prepared to double the railway line in the Western Ghat section, as this could improve cargo and passenger traffic on the network, he says.

Cargoes handled at the railway marshalling yard at NMP include edible oils, coking coal, thermal coal, finished fertilizers, and iron ore pellets, fines and lumps. The total freight handling of the Palakkad division stood at 7.34 million tonnes during 2025-26.

DELIVERING COOPERATION

Are kirana clusters the future of inclusive retail?



R SATHYANARAYANAN

Offering speed and convenience, quick commerce is rapidly transforming the urban retail landscape through agile logistics. Meanwhile, in a sign of growing anxiety in the grassroots retail community, the Khasi Hills Autonomous District Council of Meghalaya has reportedly denied quick commerce platform Blinkit a trading licence over fears it might endanger livelihoods dependent on the region's 4,000 grocery stores (kiranas).

Though a valid concern, the solution need not necessarily be to stop the advance of platforms but to build new capabilities for kiranas to achieve successful coexistence. Can cooperation offer kiranas a practical logistics response?

A cooperative kirana delivery cluster experiment in hyperlocal logistics could be imagined with around 20 kirana store members within a 3 km radius in a dense neighbourhood. Each member store shall remain independent with its customer base, pricing, credit relationships and individual identity intact, while collaborating on ordering and delivery. Through WhatsApp or another customised app, a customer selects their regular store and places an order. If that store cannot fulfil it, then the order will move to another member store in the cluster, based on transparent rules and protocols. The retailer association will employ a shared pool of delivery personnel and vehicles to deliver orders within 20-30 minutes.

The cluster development model designed by the United Nations Industrial Development Organization for small and medium enterprises (SMEs) offers useful lessons. The issue with SMEs is not just the size but also the isolation. A small store cannot individually invest in technology, logistics systems and professional service design. But a 'coopetition cluster' — rival companies competing and cooperating simultaneously to grow the overall market — allows them to share capabilities while remaining independent.

The Trichy engineering cluster is an exciting example. Years ago, the micro, small and medium enterprises (MSME) located near public sector company BHEL formed the BHEL Small Scale Industries Association. They went on to create the BHEL Industrial Development and Service Society to collectively purchase and distribute consumables to members at competitive rates.

At a global level, the Associated Wholesale Grocers, Wakefern and SPAR are examples of how a retailer cooperative model can facilitate common pro-



USP. Neighbourhood grocery stores thrive on customer relationships BLOOMBERG

curement, distribution and technology.

While pool procurement for kiranas may appear far-fetched at the moment, they can explore backend cooperation for services such as deliveries.

Direct-to-driver ride-hailing initiatives, such as Namma Yatri, provide inspiration for the ordering layer. The kirana-cluster utility could start with WhatsApp- or a light app-based ordering with a store dashboard and UPI-based payments. As the pilot matures, an integration with Nirmat Bharat's digital commerce ecosystem could be explored.

Academia can serve as a neutral cluster anchor for area mapping, designing order fulfilment rules, technology development, rider training and service monitoring. The policy environment, too, may need to become more enabling. The National Co-operative Development Corporation (Amendment) Bill, 2026, proposes wider channels for financing cooperative development, including through entities providing infrastructure, technology and other services to cooperatives.

While technology does its part, kiranas should not forget their strength in customer relationships. The *maligai kadai annachi* or *kirane-wale bhaiya* know the neighbourhood households and their individual preferences well, which can remain a competitive advantage.

The model could certainly face practical hurdles such as inventory visibility, fair order allocation, governance issues, off-peak utilisation of delivery executives, payment settlement and customer ownership. Its commercial viability, too, needs to be established first. Even so, it is a neighbourhood-level experiment worth discussing, adapting and testing as part of a more inclusive retail future.

The writer is Sundram Fasteners Associate Professor of Marketing, IFMR Graduate School of Business at Krea University

MY FIVE.

SAURABH AGARWAL
Director and CEO, HAB
Pharmaceuticals and
Research Ltd



Run on Marine Drive, Hyrox training and blue lights out

- 1 Marine Drive run: My day usually starts at 5 am. Two glasses of warm water, a banana, and I'm out for a run on Marine Drive or a gym session. The sea at that hour is the calmest meeting I'll have all day.
- 2 Hybrid athlete: I'm into hybrid fitness, a balance of endurance and strength. I eat for both. Breakfast is fruit, eggs and a smoothie. Lunch and dinner are balanced plates of carbs, protein and fibre.
- 3 Training partner: Two days of running, two for Hyrox training, one for weights and one rest day. My wife Gayatri and I race Hyrox doubles together, so fitness becomes a shared goal. The rest day is non-negotiable.
- 4 Serious recovery: Intense training needs serious recovery. I track my recovery score, HRV, resting heart rate and VO2 max on Whoop — to know when to push and when to back off. NormaTec compression, deep tissue massages and cryotherapy are regulars in my week.
- 5 Blue lights out: Thirty minutes before bed, screens go off, blue light blocking glasses go on to cut the glare from indoor lighting. Am in bed before 10, because 5 am arrives quickly.

CAPSULE.

China, South Africa tie-up

The Southern African Development Community (SADC), UNAIDS, the Chinese government and the Global Health



Innovation Institute (GHI) have agreed to advance collaboration on regional pharmaceutical manufacturing, following a joint visit to production facilities in

Johannesburg, Durban and Cape Town in South Africa, a note from UNAIDS said.

The delegation assessed existing capabilities in pharmaceutical and vaccine manufacturing, biotechnology, genomics and diagnostics, and identified priority areas to help Southern African countries expand the domestic production of quality-assured medicines and health technologies.

The collaboration aims to reduce reliance on imports, strengthen health security and improve timely access to essential health products across the region. Partnering with China offers an opportunity to connect technology, expertise, investment and manufacturing capabilities with this growing regional capacity, it added.

The collaboration aligns with the United Nations' 'Political Declaration on HIV/AIDS,' which calls for greater health sovereignty and strengthened cooperation to advance local and regional production of HIV health technologies as part of efforts to end AIDS as a public health threat by 2030. It also comes at an important moment following the UN Day for South-South Cooperation.

COMING UP.

Joining hands to beat rabies

The theme for World Rabies Day (September 28) this year is 'Stronger together', highlighting the fact that rabies elimination is possible only when we work together. "Rabies elimination depends on connected systems. Stronger partnerships create stronger outcomes," said the World Health Organization, adding that rabies is a vaccine-preventable, zoonotic, viral disease.

CARE WITH CONSENT

‘Break-glass’ allowance: How safe is personal health data?

DOS AND DON'TS. Medical bodies can now access patient data in exceptional cases, but with caveats

Vallari Sanzgiri

Consider a situation that confronts medical practitioners, more often than not, in a hospital emergency room — an unconscious, unaccompanied patient is wheeled in, who may possibly require insulin, for example.

Unlike in the past, health professionals and medical institutions can now access patient data in such exceptional situations under the Digital Personal Data Protection Act. But this special “break-glass” allowance comes with caveats, including informing the patient of this forced access, otherwise known as a ‘breach notification’. In fact, this provision covers all health data, including from diagnostic labs, where one's blood and other granular data is collected, besides specialist interactions where a patient may not comprehend how their health details may be processed.

Essentially, the DPDP Act mandates every entity processing personal information to explain how they are processing patient information, what is included in the patient information, how long this information will remain with the entity, and who is allowed to access this information. Failure to disclose these details, enforce data safeguards or even inform a patient of a data breach makes the health entity liable to a whopping fine of ₹250 crore.

The Act does come as a huge relief to patients across India — since personal health data is coveted by insurance companies, private entities and, at times, even employers who may alter their employment contracts based on the employee's health condition. What is left wanting are granular rules that further protect patients, even as they help healthcare institutions un-



ISTOCK

derstand data protection and governance in various use-cases.

“Patient privacy moves from something healthcare institutions cared about to something they will increasingly have to demonstrate. Health data has always been among the most sensitive information a healthcare provider holds. The sector has long operated under obligations spread across medical regulations, sector-specific laws and frameworks. DPDP changes the accountability layer,” explains Malcolm Gomes, Chief Operating Officer, IDfy, and Head of Privy by IDfy (which assists enterprises with DPDP compliance).

CONSENT MECHANISMS

Despite the consent-based approach, health data under the DPDP Act does not carry a separate label of ‘sensitive personal data’. Such a categorisation would have ensured additional privacy and safeguards such as “opt-in” consent mechanisms, explicit prohibition on using the data for employment purposes,

and pushing healthcare entities to undertake data protection impact assessment surveys. Further, the label would have placed restrictions on continuous collection of health data.

By clubbing health data with general personal information, the Act misses offering additional safeguards against potential misuse in the form of discrimination, stigma or loss of autonomy, says Shivangi Rai, Deputy Coordinator and Lead (technology, health and society), Centre for Health Equity, Law and Policy.

DATA MOVEMENT

Beyond categorisation, health institutions also need to understand how to safeguard patient data that moves across the system. A patient is bound to approach multiple doctors, clinics and hospitals when seeking treatment. The Act does not explain how these separate entities should coordinate with each other. Healthcare organisations also need to understand how the Act will work alongside existing clin-

ical, insurance, record retention and digital health requirements.

“Questions around retention, emergency use, research and secondary use of health data will, therefore, require careful governance rather than a one-size-fits-all approach. Children's data introduces another layer of complexity, particularly around verifiable parental consent,” says Gomes.

FOREIGNERS' TREATMENT

The entry of foreign patients further complicates these processes. A foreign patient will have to share old treatment records and insurance details with an Indian hospital before travelling for surgery. Post surgery, the hospital must share treatment updates, discharge summaries and billing records with the insurance company.

While the current law allows this, it also mentions “no transfer” or “restricted” countries, where information cannot travel.

“As of September 2026, no country has been notified as a restricted destination. However, the provision is part of the DPDP framework, whose substantive implementation is being phased in. Organisations should not treat the absence of a notified blacklist as the end of their cross-border compliance analysis,” says Rishi Agrawal, CEO and co-founder, Teamlease Regtech.

India's medical tourism industry is projected to reach \$16.21 billion by 2030 at a CAGR of 13.23 per cent, drawing over 4.5 lakh foreign patients annually, according to a Deloitte-Confederation of Indian Industry report.

Agrawal also differentiates between the law being enacted and enforced.

Notified in 2023, the Act and its related rules are being enforced in a phased manner. The watchdog (Data Protection Board of India) has also been established. In eight months, the law and its penalties will kick in.

FROM THE REGULATOR

Blueprint for AI use in the healthcare sector

The UK's National Commission into the Regulation of AI in Healthcare has published its recommendations on how that country can regulate AI in healthcare to ensure it remains safe, keeps pace with innovation and continues to earn people's trust.

AI is already used in the NHS to spot, early on, serious health problems such as strokes and skin cancer, and free up clinicians' time through voice-enabled technologies. The commission's recommendations aim to help more patients benefit safely and quickly, by supporting the UK to become the best place for innovators to build and safely test AI, the best place for healthcare professionals to use it and, most importantly, the best place for patients to engage with it in their care.

The independent commission was established by the Medicines and Healthcare products Regulatory Agency (MHRA) in September 2025 and includes diverse expertise across health, regulation and AI, including international contributors from the US and Singapore.

It gathered evidence over the course of a year from more than 12,000 people, including patients, the public, clinicians, healthcare leaders, industry and technology developers, making it the largest engagement of its kind in the UK on the regulation of healthcare technology. Additional research by the Health Foundation provided detailed insight into the UK public's priorities and expectations for the regulation of AI in healthcare.

Taken together, the engagement found broad public support for AI in healthcare, with clear conditions: Strong safety standards, meaningful human oversight and transparency about when and how AI is being used in treatments. The commission's recommendations respond directly to these views.

It recommends that the MHRA introduce staged authorisations for new AI models, similar to ‘L-plates’ for learner drivers. New models would be deployed under close supervision and with tight guardrails, allowing them to demonstrate real-world safety and performance before being granted fuller authorisation. This staged approach is designed to give UK patients world-first access to promising new models, while ensuring risk is carefully controlled at every step.

Source: MHRA



ISTOCK

Missing link in Ayushman Bharat: Emergency care

BY INVITATION



K MADAN GOPAL
KS UPLABDH GOPAL



GOLDEN HOUR. Timely, trained care saves lives and limbs ISTOCK

Eight years after its launch, the Ayushman Bharat scheme has shown that India can create health entitlements at an unprecedented scale. More than 45 crore Ayushman cards have been issued, and 12 crore hospital admissions authorised.

The scheme has developed a wider health system architecture: The Ayushman Arogya Mandirs strengthen primary care, PM-ABHIM is adding critical-care infrastructure, and the Ayushman Bharat Digital Mission (ABDM) is building the digital backbone. The challenge is in making these building blocks work together. Nothing illustrates this gap more than a medical emergency.

A person with a serious road accident injury, heart attack, stroke, sepsis or obstetric emergency does not enter the health system in an orderly manner. What matters is a quick response, trained care before hospitalisation and timely access to appropriate treatment.

India is not starting from scratch. The 112 Emergency Response Support System provides a national emergency number, while States operate ambulance services under the National Health Mission. The National Ambulance Service Operational Guidelines, 2026, aim for standardisation in despatch, triage, tracking and emergency medical technician (EMT) skills.

The recently introduced PM-RAT scheme for road accident victims provides an important facility. It enables cashless treatment of up to ₹1.5 lakh for up to seven days at designated hospitals. When life is at risk, treatment should not wait while a family arranges money or establishes insurance entitlement.

An ambulance may reach a patient without knowing which nearby hospital has the required specialist, blood, operating theatre or intensive-care bed. A patient may be taken to the nearest hospital only to be referred again, with little information

about what happened during transport. Critical care blocks have limited value if the referral system cannot bring in the right patient in time. For families, these are lost minutes.

The 112 platform and State ambulance systems could form the backbone of an integrated emergency response, with command centres tracking ambulances and hospital capacity. A severely injured person should be taken not simply to the nearest hospital but one capable of managing that injury. Ambulances must become an extension of clinical care, with trained EMTs.

At the hospital, a critically ill patient should receive immediate stabilisation without waiting for a deposit, identity document or insurance approval. A transparent financing mechanism could enable participating private hospitals to provide initial stabilisation without shifting the cost to families. PM-RAT provides a useful pathway.

District and select sub-district hospitals should also form part of a graded emergency care network with defined capabilities and referral pathways. ABDM can help create this continuum by supporting pre-arrival information, and referral and transfer summaries with privacy and consent safeguards.

The journey also begins and ends in primary care. Improved control of hypertension and diabetes can, in fact, prevent some emergencies, while rehabilitation and follow-up must continue after discharge.

Dr Madan Gopal is a former senior health consultant with NITI Aayog, and Dr KS Uplabdh Gopal is a health professional. Views are personal

Wellness-centric care with nutraceuticals

Deciphering the right dose of alternatives to illness-centric, drugs-based healthcare

DB Anantha Narayana

A neighbour walked into my house with a bottle of melatonin tablets, labelled “nutraceuticals”, and wanted to know if he could give it to a relative who had just flown in from abroad.

I gave my “go ahead” as melatonin is not a drug but a hormone produced in the pineal gland of the human brain for sleep and sleep cycle management. It is also manufactured in factories now as a nutraceutical, made available as tablets for convenience, and helps fight jet lag and promote good sleep.

HISTORY OF USAGE

Dr Stephen DE Felice, in 1989, coined the term “nutraceutical” by combining “nutrition” and “pharmaceuticals”. And unlike melatonin, a number of other compounds useful for



LEAFY CURE. Most nutraceuticals have their origin in herbs ISTOCK

health are sourced from herbs.

In fact, herbs have a long history of human consumption (HAHC). In the Indian context, many herbs have a documented long history of safe use (HoSU) in traditional

knowledge texts. These herbs are processed for consumption through steps such as infusion and decoction, and taken together with milk and so on.

In the last few decades, the industry has converted such

herbal extracts into powders through spray drying and other technologies. They are then sold as powders, granules, capsules, tablets, gels or value-added food stuff, with specified usage for convenience.

Such products help consumers move from illness-centric healthcare through drugs to wellness-centric healthcare.

REGULATORY OVERSIGHT

The products are generally called food or health supplements and are regulated under the FSSAI's ‘Food Safety and Standards Act and Rules’.

A warm cup of tea is often consumed to sleep better. L-theanine in tea was seen, in human studies, to reduce the lag-time in falling asleep.

A 200 mg L-theanine product in the evening is useful to sleep better. Curcuminoids from turmeric; zeaxanthin and lutein from marigold petals;

resveratrol from the skin of grapes; hesperidin and quercetin from lemon peel are a few examples of nutraceuticals.

Herbs have compounds in them, which are the basis for their health benefits. Scientists have isolated and produced many such compounds from herbs in their pure form. Most nutraceuticals have their origins in one or more herbs.

Scientist Dr Vishweshwaraiah Prakash, former Director of the Central Food Technology Research Institute, Mysuru, coined the term “herbaceuticals” to promote the appropriate use of herbs and compounds for their health benefits. Their consumption, with the right dosage information, is, in fact, the best prescription.

The writer is a regulatory expert, pharmacist and former Chairman, Expert Committee on Nutraceuticals, FSSAI. Views are personal

QUICKLY.
US and China begin talks
ahead of Xi-Trump meet



US and Chinese officials began talks in New York to lay the groundwork for a high-profile summit between Presidents Donald Trump and Xi Jinping, as the world's two biggest economies seek to maintain and potentially extend a fragile trade truce. The negotiators, led by US Treasury Secretary Scott Bessent and China's Vice Premier He Lifeng, are expected to discuss issues spanning trade and investment, artificial intelligence and the Iran war. He, US Trade Representative Jamieson Greer and Li Chenggang, China's international trade negotiator, entered the JPMorgan Chase & Co. offices for the talks. Bessent and He last met near Seoul four months ago, just days before Trump visited Beijing in May. On Thursday, Trump will host Xi at the White House in the Chinese leader's first visit to Washington in over a decade. BLOOMBERG

DRDO's 30 kW directed-energy weapon to be ready within a year

COUNTERING DRONES. It will issue an EoI in 6 months to select a production partner, says BK Das

Dalip Singh
New Delhi

The Defence Research and Development Organisation's (DRDO's) first-of-production model (FOPM) for the 30 kW-class directed-energy weapon (DEW), a strong arm of the 'Sudarshan Chakra' comprehensive air defence system announced by Prime Minister Narendra Modi earlier, will be available within a year to counter swarm drones and unmanned aerial vehicles (UAVs). The truck-mounted DEW uses electromagnetic energy for destroying targets either by way of soft kill or hard kill. "Having demonstrated the technology, DRDO will issue an expression of interest (EoI) in the next six months to select a production partner for the 30 kW high energy laser beam weapon system," BK Das,

GOING HI-TECH

- The 30 kW directed-energy weapon will counter swarm drones
- The truck-mounted DEW uses electromagnetic energy for destroying targets either by way of soft kill or hard kill
- India had deployed the DRDO's 2kW DEW along the Line of Control during Operation Sindoor


BK Das

Director General of the Bengaluru-based DRDO lab Electronics and Communication Systems (ECS), said in an exclusive interaction with *businessline*. The technology trial was carried out last year. "Within a year, we hope to see the FOPM. It will undergo further tests and trials to ascertain range and other parameters desired by users," he said, explaining the cycle from the drawing board to the shop floor.

'OODA LOOP'
This will be part of the the 'OODA loop', a decision-making military framework that stands for 'Observe, Orient, Decide and Act.' The DEW is emerging as a strong arm of the multiple approach of neutralisation where the choice of weapon, be it kinetic or non-kinetic, is conditioned on the incoming projectiles, as per the DG of ECS.

On the overarching coun-

trywide air defence system Sudarshan Chakra, Das said, "It's a great vision by the Prime Minister and the country needs it. We have most of the ingredients in some form or the other in our ecosystem. We need to integrate all of them for an effective layered defence system." In the entire push, private industry, MSMEs and start-ups are deeply involved. This is the third in the series of DEWs developed indigenously by DRDO, orbiting India into a select few club of countries like the US, China, Russia, Israel, the UK and Germany that possess such capabilities, gaining relevance in the modern battlefield defined by the economics of warfare.

2kW DEW
India had deployed DRDO's 2kW DEW along the Line of Control (LoC) in Operation Sindoor as part of its multi-

layered inner-tier air defence network to counter asymmetric aerial threats from Pakistan over a range of up to 1 km. The DRDO has selected multiple production partners, including Bharat Electronics Ltd and Adani Defence & Aerospace. The more potent version of 10 kW DEW can engage drones, UAVs and sensors within a range of 1.5 km to 2 km. The Defence Acquisition Council, chaired by Defence Minister Rajnath Singh, had issued Acceptance of Necessity (in principle approval) for 10 to 12 numbers of 10 kW DEW, which will be subsumed into the Integrated Drone Detection and Interdiction System Mark-2. Now, the development-cum-production partner L&T has to come up with an FOPM of 10 kW class DEW and subsequent to that it will undergo further trials, the leading scientist stated.

Tata Sons-Trusts battle headed for legal showdown

Our Bureau
Mumbai



The escalating dispute between Tata Sons and Tata Trusts over the reappointment of Chairman N Chandrasekaran increasingly appears headed for a legal showdown, with two of the country's most prominent legal minds now associated with opposing sides. Senior advocate Harish Salve is understood to be advising the Chandrasekaran-led Tata Sons, while senior advocate Abhishek Manu Singhvi has entered the fray on behalf of the Tata Trusts. The disagreement stems from Tata Trusts' challenge to the validity of the September 17 board resolution that approved Chandrasekaran's reappointment for a further five-year term.

the dispute with "sadness and regret", noting his association with the late Ratan Tata and his longstanding relationship with key individuals on both sides. He argued that the fundamental rights of shareholder-owners could not be nullified in the manner alleged by the Trusts.

TRUSTS REJECT REMEDY
The sharp exchange follows Tata Trusts' assertion that the board resolution failed to secure the affirmative support of a majority of Trust-nominated directors as required under the AoA and, therefore, could not be validated through a casting vote. The Trusts have also argued that listing Tata Sons should not be viewed as a governance solution, noting that the company already follows several governance practices associated with listed entities. The dispute has drawn in members of the wider Tata family. In an email dated September 17, Mehli Mistry told Tata Trusts officials that he was "saddened" by the litigation, describing it as unprecedented in over 150 years. He also stated that the Tata Education & Development Trust would not bear any legal expenses arising from the dispute and asked for his communication to be formally recorded at the Trust's next board meeting on November 20.

CYRUS MISTRY RULING
Singhvi on Sunday publicly backed the Trusts' position, invoking the Supreme Court's verdict in the Tata Sons-Cyrus Mistry case, where the court upheld the special rights granted to the Tata Trusts under the company's Articles of Association (AoA). In a post on X, Singhvi said the judgment clearly recognised the primacy of Tata Trusts in their relationship with Tata Sons and upheld the special articles designed to protect that position. "Unfortunately, in the absence of collegiality and conviviality, these and many other related issues can only have legal solutions," Singhvi wrote, adding that the Supreme Court judgment and the fiduciary obligations owed to millions of Indians appeared to have been "selectively forgotten". In a separate statement, Singhvi said he was entering

Adani seeks nod for US pre-clearance at Navi Mumbai airport

Rohit Vaid
New Delhi

Airport operator Adani Airport Holdings Limited (AAHL) has sought the Centre's approval to set up a facility for pre-clearance of US immigration and customs formalities for passengers travelling on direct US-bound flights from the Navi Mumbai International Airport (NMIA). Speaking to *businessline*, industry sources said the proposal had been submitted to the Ministry of Civil Aviation (MoCA). The proposed facility

would enable passengers to undergo US immigration and customs checks before boarding their flights. Besides, the facility would benefit passengers connecting to onward flights within the US by allowing them to bypass routine immigration and customs queues on arrival. When contacted, AAHL did not respond to queries from *businessline*.

PROJECT COST
Meanwhile, details of the implementation timeline, infrastructure and final investment requirement were not immediately available.

However, per sources, the proposed project at Navi Mumbai is estimated to cost over ₹100 crore. Further, an agreement as well as arrangements between the Indian and US authorities would be required to establish the facility. The US immigration and customs pre-clearance is administered by the US Customs and Border Protection (CBP). Under the pre-clearance model, CBP personnel conduct inspections at designated overseas airports before passengers board US-bound aircraft. At present, Abu Dhabi's



Zayed International Airport and Dublin Airport host US pre-clearance facilities. According to a senior airline executive, such a facility could make the Navi Mumbai airport attractive not only to US-bound passengers originating in India but also to international transit passengers travelling to the US.

The proposal comes as NMIA seeks to expand its international connectivity. Recently, the Airports Economic Regulatory Authority of India (AERA) approved the airport operator's proposal to revise eligibility criteria for tariff incentives available to airlines launching international passenger routes from the NMIA.

AIRLINE INCENTIVES
Notably, under the framework, airlines launching eligible new international passenger routes of up to 5,000 km receive a 100 per cent concession on landing charges in the first year and

50 per cent in the second year. Subsequently, the concession would drop to 50 per cent in the second year and 25 per cent in the third year. For routes exceeding 5,000 km, the concession is 100 per cent in the first year, 50 per cent in the second year and 25 per cent in the third year. Nevertheless, the commercial opportunity for Navi Mumbai would depend on airlines offering direct US services and connecting schedules that allow passengers from other Indian cities or overseas markets to transfer through the airport.



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